

Information Leakage Prior to SEC Form Filings: Evidence from *TAQ**

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Abstract

Using *TAQ* data around trading-hour SEC filings from 2003 to 2024, we document a pre-announcement drift across EDGAR acceptance timestamps: when firms experience significant cash-flow news, 30-minute pre-acceptance returns positively predict 30-minute post-acceptance returns. Momentum, investor attention, recent firm-specific news, and contemporaneous market-wide movements do not explain the pattern; and this pattern would disappear if genuine filing date-times are replaced by placebo with random date-times. Extreme pre-acceptance returns coincide with elevated abnormal volume, informed-trading risk, and directional order flow, and the drift strengthens with SEC enforcement caseload. Our evidence is consistent with filing information reaching prices before public dissemination.

Keywords: Information Leakage, Informed Trading, Price Discovery, SEC Filings, EDGAR

JEL Classification: G10, G12

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I have nothing to disclose.

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1. Introduction

Corporate information can enter market prices before its formal disclosure, but existing evidence is concentrated in particular announcement types and leakage channels. We ask how broadly this phenomenon extends across SEC filings and whether it is visible within minutes of release. Using trading-hour filings and exact EDGAR acceptance timestamps, we document a high-frequency pre-announcement drift: returns before acceptance positively predict returns after acceptance. The relation appears across major filing categories, suggesting that pre-release price discovery is not confined to the specialized settings studied previously.

This pre-announcement drift differs from the post-earnings-announcement drift of [Bernard and Thomas \(1989\)](#). Post-earnings-announcement drift is the continuation of returns in the direction of a disclosed earnings surprise over the weeks after the announcement. The pre-announcement drift is instead the continuation, within 30 minutes of acceptance, of a price move that begins before the filing is public. It also differs from the pre-FOMC announcement drift of [Lucca and Moench \(2015\)](#), which is an average excess return before scheduled announcements rather than the continuation of a price move across the release time. Throughout the paper, we use the terms *continuation* and *pre-announcement drift* interchangeably, to denote the main empirical pattern that we document.

Existing studies document information leakage around specific events, such as sovereign downgrades and syndicated loan negotiations (e.g., [Michaelides et al. \(2015\)](#), [Ivashina and Sun \(2011\)](#)). We examine whether such leakage extends across the full range of SEC filings.

Using intraday data during trading hours allows us to measure pre-release price discovery on a much narrower clock than prior work. Daily or monthly event windows can confound the impact of the filing on market outcomes with other firm news, and poor signal-to-noise ratio at longer horizons makes price-based detection of information leakage difficult. By focusing on narrow windows immediately around a precisely timestamped disclosure, we reduce these confounds and ask whether information contained in pre-acceptance prices predicts the market’s subsequent response. Because the pre-acceptance return is model-free and uses only prices observed before acceptance, it can be computed in real time. Trading-hour data also allow us to measure informed-trading risk before acceptance with the QID^{res} measure by [Barardehi, Dixon, and Liu \(2026\)](#).

We study SEC filings accepted during regular trading hours from September 2003 through December 2024. EDGAR acceptance provides a precise event clock: accepted filings are disseminated within two minutes.¹ We use the acceptance timestamp as event time and measure returns over the 30 minutes before and after it.

Certain ownership and insider-trading forms require separate treatment. Forms 3, 4, 5, SC 13D, and SC 13G disclose transactions or position accumulation that has already occurred, so their pre-filing price movement can reflect the filer’s own completed trading and position build-up rather than information in the forthcoming disclosure. Leakage of these forthcoming disclosures can also be counterproductive to the filing parties because it invites front-running. We therefore exclude these forms and their amendments from our baseline sample, which we call EXCL.

Leakage also requires an incentive, and the incentive exists only when the forthcoming disclosure has significant information content. A filing that confirms what the market already expects offers little to trade on before its release. Filings carry significant information mainly when the firm has experienced major updates to its fundamentals, which should be reflected in its operating cash-flow (hereafter, cash-flow). We therefore focus on filings made in firm-quarters in which the firm’s realized operating cash-flow deviates substantially from an econometric forecast based on firm-specific and industry-level characteristics. Each filing is assigned to the quarter that ends with the firm’s next earnings announcement, and we retain filings in quarters whose absolute cash-flow forecast error exceeds the median, which we call informative firm-quarters. We refer to this restriction as the cash-flow filter.

In EXCL (i.e., the baseline sample), filings in the highest quintile of pre-acceptance returns subsequently outperform those in the lowest quintile by 8.86 basis points ($t = 6.78$; $t = 6.71$ with standard errors clustered two-way on firm and month), and post-acceptance returns increase monotonically across pre-acceptance-return quintiles. In the all-form-type sample, which retains the ownership forms, the spread remains highly significant at 4.18 basis points ($t = 5.22$). Our main empirical findings remain statistically significant without the cash-flow filter. Table A.1 in the Appendix reports them for comparison.

This return continuation is notable because half-hour stock returns tend to reverse

¹ Per the [EDGAR public dissemination service technical specification](#): “the acceptance and dissemination process usually takes no longer than two minutes from the receipt of submission by EDGAR.”

unconditionally (Heston, Korajczyk, and Sadka (2010)). Absent filing-specific information in pre-acceptance prices, the natural benchmark is therefore zero or a negative correlation (i.e., reversal), rather than a positive correlation (i.e., return continuation) across the acceptance timestamp.

The relation is not concentrated in a single filing class. It appears among both scheduled and unscheduled filings, in Form 8-K, and in Form 10-Q. It also remains significant after excluding Forms 8-K, 10-K, and 10-Q. The breadth of these results makes it unlikely that the baseline pattern is generated by one particular disclosure category.

We next examine several non-leakage explanations. The relation survives after controlling for EDGAR-based investor attention, and variables that capture past returns; and results also remain after controlling for market adjustment using the contemporaneous S&P 500 return, and the removal of filings preceded by *RavenPack* firm-specific news in the six days before the filing or earlier on the filing date. These tests do not eliminate every possible information channel, but they make investor attention, momentum, broad market movements, and observed recent news unlikely to account for the baseline relation.

The predictive information is concentrated in the half hour before acceptance. Controlling for various lagged returns, ranging from the preceding overnight return to to one year trend, leaves the coefficient on the 30-minute pre-acceptance return essentially unchanged, at 0.0658 against 0.0661 without controls. Sorting filings on the previous day’s or week’s return instead of the 30-minute pre-acceptance return produces reversal rather than continuation, with spreads of -4.85 basis points ($t = -3.95$) and -3.43 basis points ($t = -2.88$). The pre-announcement drift we document at the 30-minute time interval is therefore distinct from lower-frequency return trend.

The continuation is also specific to filing timestamps. Among 98,374 random firm–day–time placebo events, drawn from the same informative firm-quarters, the top-minus-bottom spread is statistically indistinguishable from zero; the filing-minus-placebo difference (i.e., between the genuine sample versus a randomly constructed placebo) is 8.14 basis points ($t = 4.57$). Comparable half-hour returns at arbitrary times therefore do not display the pattern observed around EDGAR acceptance. The baseline result is unchanged under 90-minute event windows, alternative event-price definitions, and quote-midpoint returns. The panel regression results remain robust under filing-date fixed effects, with or without firm

fixed effects.

Known technical and criminal leakage episodes do not account for the baseline result. Bolandnazar et al. (2020) document an SEC dissemination-system defect that gave some users seconds-to-minutes early access to filings, which was largely addressed and fixed by March 2015.² This defect operates on a far shorter interval than our 30-minute windows. Separately, regulators prosecuted trading on hacked corporate announcements through September 2020.³ The pre-announcement drift remains significant in samples beginning after each date, indicating that the phenomenon extends beyond these documented episodes. Barger, Clifford, and Qiu (2025) identify filing agents as a potential leakage channel around after-hours Form 8-K releases. We show that the pre-announcement drift is significant in both the agent-filed and the self-filed subsamples, and beyond just Form 8-K.

Several patterns support the leakage interpretation of this drift. First, others measures indicate informed trading before acceptance. Both QID^{res} —which is a new proxy for measuring informed-trading by Barardehi, Dixon, and Liu (2026)—and abnormal volume, are highest in the extreme pre-acceptance-return quintiles, and return continuation is strongest where these measures are elevated. Lee–Ready directional order flow (Lee and Ready (1991)) is also monotonic in pre-acceptance return: buyer-initiated volume rises and seller-initiated volume falls across pre-acceptance-return quintiles. Thus, the pre-filing price movement is accompanied by trading intensity and order flow aligned with the subsequent price response.

Second, the pre-announcement drift varies with the SEC’s enforcement caseload. Relative to its level at the mean caseload, a one-standard-deviation increase in the SEC’s open enforcement caseload nearly doubles the continuation coefficient. The intuition is that facilitators of information leakage balance the expected profits from informed trading against the risk of detection. If the SEC’s enforcement capacity is fixed in the short run, a higher caseload leaves the SEC less capacity to pursue firms that are not already under investigation. When the SEC is occupied with ongoing cases, facilitators of information leakage therefore face a lower risk of detection and return continuation phenomenon is more prominently observed.

² Per Bolandnazar et al. (2020): “[t]he median delay is 26 seconds.”

³ See the press releases of the U.S. Department of Justice and the U.S. Securities and Exchange Commission (2015 and 2019).

The pre-acceptance signal is economically meaningful. Following the methodology of [Bernile, Hu, and Tang \(2016\)](#), with reasonable assumptions, we estimate and find significant profits for informed trading in the pre-acceptance period. Additionally, because the pre-acceptance return is observed before the trade, we can also measure how much information remains unpriced at acceptance with a real-time alpha-generating trading rule without being complicated the look-ahead bias. As we aim to construct a real-time trading rule, we do not invoke any future information. Hence, the trading rule uses prior-year cutoffs, trades every EXCL filing without applying the cash-flow filter, enters three minutes after acceptance, and trades only on the extreme pre-acceptance returns. At quote midpoints, the rule earns a monthly five-factor alpha of 1.81% ($t = 3.47$); at transaction prices, the alpha is 1.62% ($t = 2.85$). We interpret this exercise as evidence that the information remains unpriced at filing acceptance, rather than as the performance of a fully implementable net-of-cost trading strategy.

Taken together, our results establish broad pre-acceptance price discovery around trading-hour SEC filings. Our contribution is to show that this phenomenon appears across a wide range of filing types, survives alternative explanations, and is accompanied by directional and informed-trading proxies, and varies with SEC enforcement caseload. The phenomenon loses its directionality if we use past 1-day return instead of the 30-minute pre-acceptance return. The placebo tests also point to the direction that the phenomenon is not an artifact. The joint evidence is consistent with filing-related information reaching some market participants before public dissemination, although our design does not identify the source or transmission channel.

The rest of the paper is organized as follows. [Section 2](#) relates our findings to the existing literature. [Section 3](#) describes the data and empirical design. [Section 4](#) presents the main results. [Section 5](#) presents robustness tests and results. [Section 6](#) presents evidence on the leakage interpretation. [Section 7](#) assesses the economic significance. [Section 8](#) concludes.

2. Related Literature

Our work relates to three strands of literature: information leakage before announcements, high-frequency return dynamics, and the market’s response to news.

Information leakage before announcements. Busse and Green (2002) study the “The Morning Call” and “Midday Call” segments on CNBC TV, and find significant price run-up in the *five-minute* interval prior to positive reports, the timing is very close to airing and “[t]he run up in prices before the report is consistent with some traders becoming aware of the information before the segment airs.” Prior works also document price and order-flow movements before a range of public information events, including FOMC announcements (Bernile, Hu, and Tang (2016)), sovereign rating changes (Michaelides et al. (2015)), takeover bids and earnings announcements (Del Guercio, Odders-White, and Ready (2017)), takeovers (Augustin, Brenner, and Subrahmanyam (2019)), share repurchases (Qing (2016)), analyst downgrades (Christophe, Ferri, and Hsieh (2010)), syndicated-loan information (Ivashina and Sun (2011)), and specific set of events such as unscheduled corporate announcements (M&As, SEOs, and dividend initiations) and pre-scheduled earnings announcements (Brennan, Huh, and Subrahmanyam (2018)). Other work identifies leakage channels including illegal tipping networks (Ahern (2017)), hacking of earnings (Akey, Grégoire, and Martineau (2022)), and defects in SEC dissemination systems (Bolandnazar et al. (2020)). Legal and enforcement risk also shapes how informed investors trade (Crimmins (2013), Kacperczyk and Pagnotta (2024)).

These studies typically identify pre-release trading within a particular announcement topic or a single form-type. Rogers, Skinner, and Zechman (2017) study the SEC dissemination process for Form 4 filings and show that paying PDS subscribers historically received many filings before those filings appeared on the EDGAR website.⁴ Barger, Clifford, and Qiu (2025) identify filing agents as a potential leakage channel around after-hours Form 8-K releases. Our setting differs from both: we study price dynamics before the EDGAR acceptance timestamp across a broad set of trading-hour filings and exclude Forms 3, 4, 5, SC 13D, and SC 13G from the baseline sample. This design allows us to study whether pre-release price discovery is a broader phenomenon, rather than one confined to a specific

⁴ Using 22 months of PDS data between March 2012 and December 2013, Rogers, Skinner, and Zechman (2017) find that “[t]he average period of private advantage is about 10 seconds for the full sample and 18 seconds for the subsample of filings where the PDS feeds are before the SEC posting, a relatively long time in the world of high frequency trading.” This private advantage is a technical one that arises after acceptance, in the SEC’s dissemination of accepted filings to PDS subscribers and to the EDGAR website. We instead study price discovery before the EDGAR acceptance timestamp. Like the dissemination-system defect documented by Bollandnazar et al. (2020), an advantage of 10 to 18 seconds operates on a far shorter interval than our 30-minute windows.

filing type. Our setup allows for an array of empirical techniques, and results suggest the presence of information leakage beyond Form 8-K, and in self-filings.

High-frequency return dynamics. A second strand studies how returns evolve within the trading day. [Heston, Korajczyk, and Sadka \(2010\)](#) show that half-hour returns tend to reverse, which sets the unconditional benchmark in Section 3.4, and [Lou, Polk, and Skouras \(2019\)](#) and [Bogousslavsky \(2021\)](#) show that the intraday and overnight components of returns behave very differently across the cross-section. In contrast to these general return pattern, and the null hypothesis of return reversal, we instead find that half-hour returns continue rather than reverse around EDGAR acceptance timestamps, and the continuation pattern is absent at randomly drawn firm-date-times.

Market response to news. A third literature studies how quickly prices incorporate public information. Prices underreact to earnings announcements ([Bernard and Thomas \(1989\)](#)) and to firm news more generally at high frequency ([Jiang, Li, and Wang \(2021\)](#)), with slower adjustment when institutional attention is low ([Ben-Rephael, Da, and Israelsen \(2017\)](#)). Short sellers trade as news arrives ([Engelberg, Reed, and Ringgenberg \(2012\)](#)), managers time disclosures to periods of low attention ([Niessner \(2014\)](#)), and returns accrue before scheduled FOMC announcements ([Lucca and Moench \(2015\)](#)). Our results shift the event clock backward: for a broad set of SEC filings, the market response begins before the SEC acceptance of filings, and the pre-acceptance return predicts the post-acceptance return. Firm-specific *RavenPack* news in the six days before the filing or earlier on the filing date does not explain this continuation.

3. Data and Measures

3.1. Data

We obtain each filing’s CIK, accession number, form code, and acceptance timestamp from WRDS and combine these records with *CRSP*, *TAQ*, and the WRDS SEC Analytics Suite. We use the *TAQ* Monthly Product from 2003 through 2012 and the Daily Product from 2013

through 2024; the products overlap in 2013, which we use to validate the transition. Because the acceptance timestamp is available in our source only from September 2003 onward, our sample begins thence. We define event time as the EDGAR acceptance timestamp. Because acceptance precedes the PDS dissemination and website posting studied by [Rogers, Skinner, and Zechman \(2017\)](#), our *pre*-event window ends before the SEC distributes the filing through any channel. SEC technical documentation indicates that accepted filings are disseminated publicly within two minutes ([Securities and Exchange Commission \(2009\)](#)).

Data preparation. We match each filing’s CIK to a CUSIP using the WRDS SEC Analytics Suite and require the link to be valid on the event date.⁵ We restrict the sample to filings accepted during regular trading hours. In after-hours trading, volume is low, market makers seldom post firm quotes, and trading costs are four to five times those of the trading day ([Barclay and Hendershott \(2003\)](#)). Because after-hours prices are also noisier ([Barclay and Hendershott \(2003\)](#)) and the quotes are unconsolidated, return-based tests of leakage have less power outside regular trading hours. For the baseline specification, we retain filings accepted between 10:00 a.m. and 3:30 p.m. ET so that the full 30-minute pre- and post-acceptance windows fall within regular trading hours.⁶ We restrict the sample to common shares, SHRCD 10 and 11, traded on NYSE, AMEX, NASDAQ, or Arca (*CRSP* exchange codes 1–4) at the filing date. Finally, among the remaining filings, we remove filings that occur within 30 minutes of another such filing by the same firm on the same day, including a gap of exactly 30 minutes, so that a return window cannot be attributed to multiple nearby disclosures.⁷

We apply the [Holden and Jacobsen \(2014\)](#) filters to both the monthly and daily *TAQ* products. For quotes, we retain valid NBBO observations and remove abnormal quote conditions, crossed markets, spreads above five dollars, and observations with non-positive or missing prices or sizes. Moreover, for trades, we retain regular, uncorrected transactions

⁵ The event date must fall between the first and last filing dates of the CIK–CUSIP combination (CIKDATE1 and CIKDATE2).

⁶ For the 90-minute specification, we retain filings accepted between 11:00 a.m. and 2:30 p.m. ET.

⁷ If a firm files a Form 10-K at 11:00 a.m. and a Form 4 at 11:05 a.m., both filings are dropped. If it files a Form 8-K at 11:00 a.m. and a Form 10-Q at 11:45 a.m., both are kept. The same rule, with a 90-minute gap, is applied in the 90-minute sample.

with positive prices and exclude abnormal sale conditions.⁸ We use trades reported on the NYSE, NYSE American, and Nasdaq (*TAQ* exchange codes N, A, T, and Q). After matching *TAQ* observations to filing events and imposing the return-window requirements described below, 157,154 unique EXCL filings have valid 30-minute returns. Restricting these filings to informative firm-quarters, as described in Section 3.3, yields the baseline 30-minute EXCL sample of 68,569 filings. Table IA.1 in the Internet Appendix reports the complete sample-selection funnel.

3.2. Variable Construction

Event returns. We construct pre- and post-acceptance returns from *TAQ* transaction prices. Because a trade need not occur at an exact timestamp, we measure each endpoint as the mean transaction price over a three-minute interval. The three-minute width reduces missing observations and accommodates the short delay between EDGAR acceptance and dissemination while keeping endpoint construction identical across measures ([Securities and Exchange Commission \(2009\)](#)). In our baseline analysis, we require actual trades to take place in order to calculate the *starting price* and *ending price* of the pre- and post-acceptance windows, to be described in the following paragraph. This requirement focuses the analysis on filings whose stocks actually trade before and after acceptance, where leaked information would appear in executed prices, at the cost of a smaller sample. As a robustness exercise, we relax the restriction of having actual trades and use quote midpoints instead in Section 5.3, which increases our sample size.

We define the *starting price* of the pre-acceptance period P_{-30} as the mean price over the three-minute interval from 30 minutes to 27 minutes before the acceptance timestamp. We define the *just-before-acceptance* event price P_0^{pre} , or the *ending price* of the pre-acceptance period, as the mean price over the three-minute interval from three minutes to one millisecond before the acceptance timestamp. We calculate the pre-acceptance return Q_{30}^{pre} as the log of the ratio of three-minute mean prices from the window $[-30, -27)$ to the window $[-3, 0)$, in

⁸ We follow [Holden and Jacobsen \(2014\)](#) to screen quotes: valid quote conditions, a national best bid and offer indicator, and a missing symbol suffix. We also only use regular trades, that is, those with sale conditions of blank, @, E, F, I, S, or Y and a trade-correction indicator of 00 ([Federal Register](#)).

minutes relative to the acceptance timestamp:

$$Q_{30}^{pre} = \log\left(\frac{\bar{P}_{[-3,0]}}{\bar{P}_{[-30,-27]}}\right). \quad (1)$$

Q_{30}^{pre} is the return of the filing firm from 30 minutes before the acceptance timestamp up to the three minutes immediately preceding acceptance. We similarly define Q_{90}^{pre} over the 90-minute interval before the filing.

We construct the post-acceptance return for each SEC form filing in a similar manner. We define the *starting price* of the post-acceptance period as the mean price over the three-minute interval from acceptance to three minutes after, and the *ending price* as the mean price over $[27, 30)$. We calculate

$$Q_{30}^{post} = \log\left(\frac{\bar{P}_{[27,30]}}{\bar{P}_{[0,3]}}\right). \quad (2)$$

Q_{30}^{post} is the return from the three minutes immediately after acceptance to 27–30 minutes after acceptance. The pre- and post-acceptance returns therefore share no price and cover disjoint intervals. Constructing Q_{30}^{post} requires a transaction in both of its edge windows, so we drop filings with no actual trade between the acceptance timestamp and three minutes after, or between 27 and 30 minutes after. Figure 1 marks the two windows for the 30-minute case.

We similarly define Q_{90}^{post} over the 90-minute interval after the filing. We use Q_{30}^{pre} and Q_{30}^{post} throughout the main analyses. Both are model-free: each is a log return over a fixed clock-time window, computed without an expected-return benchmark and observable in real time.

Alternative return definition. With its *starting price* calculated over the $[0, 3)$ interval, the baseline post-acceptance return starts from the three minutes after acceptance. In an alternative definition, we calculate the *starting price* of the post-acceptance period over the $[-3, 0)$ interval, allowing the post-acceptance return to capture the entirety of the market’s

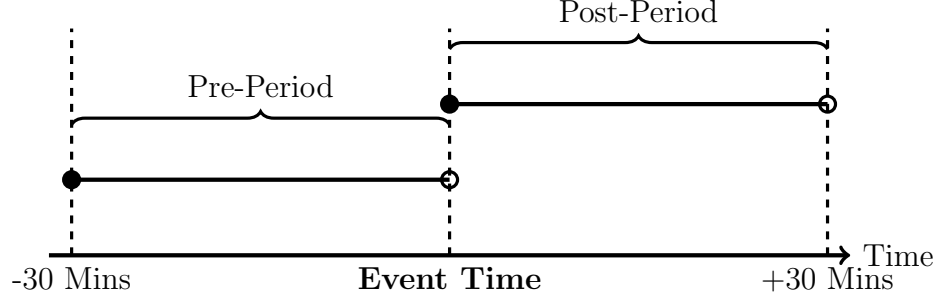


Figure 1: Definition of the pre- and post-event periods, 30-minute case

Note: The horizontal axis is time in minutes relative to the EDGAR acceptance timestamp. Filled markers denote closed endpoints and hollow markers denote open endpoints, so the pre-acceptance period is $[-30, 0)$ and the post-acceptance period is $[0, 30)$. Each return is computed from mean transaction prices over the first and last three minutes of its period, as in equations (1) and (2).

initial reaction, and keep the baseline pre-acceptance return:

$$Q_{30}^{pre,alt} = Q_{30}^{pre} = \log\left(\frac{\bar{P}_{[-3,0)}}{\bar{P}_{[-30,-27]}}\right), \quad (3)$$

$$Q_{30}^{post,alt} = \log\left(\frac{\bar{P}_{[27,30)}}{\bar{P}_{[-3,0)}}\right). \quad (4)$$

Under this definition, the pre- and post-acceptance returns share the price over the $[-3, 0)$ interval.

Form-type groups. The EXCL sample excludes Forms 3, 4, 5, SC 13D, and SC 13G and their amendments.⁹ Within EXCL, we distinguish scheduled filings, whose filing *dates* are known in advance, from unscheduled filings. Even for scheduled forms, the exact acceptance *time* remains unknown until the filing is submitted. The scheduled form types, together with their amendments, are listed and described, by type of filer, in Table A.4. The unscheduled form types are the remaining EXCL forms.

⁹ Insider trading has two distinct definitions, with implications for its legality: securities trading by a corporate insider, and, securities trading while in possession of material nonpublic information (Metrick (2018)). The federal securities laws require officers, directors, and holders of more than 10% of a class of a company's securities to report purchases, sales, and holdings on Forms 3, 4, and 5.

Pre-event trading. We use three measures of trading activity before acceptance. First, abnormal volume is the firm’s *TAQ* share volume during the 30-minute pre-acceptance window, standardized relative to its recent trading volume.¹⁰ Second, we measure informed-trading risk using the 30-minute inverse residual QID (i.e., QID^{res}) measure by Barardehi, Dixon, and Liu (2026), which is non-directional.¹¹

Third, we measure directional order flow as

$$X_{LR} = \frac{V_B - V_S}{V_B + V_S}, \quad (5)$$

where V_B and V_S are buyer- and seller-initiated share volume during the 30 minutes before acceptance, classified using Lee and Ready (1991) as implemented by Holden and Jacobsen (2014).

Control variables. To explore alternative explanations for the pre-announcement drift, we control for firm size and construct two families of control variables for the panel regressions of Section 4.4. Log firm size is the logarithm of *CRSP* market capitalization, in millions of dollars, on the filing date.

To measure investor attention, we use EDGAR server-logs over the 30 calendar days before each filing. Our three measures are total non-robot views of the firm’s filings, an indicator for any institutional view, and the number of institutional views. Following Ryans (2017), we remove accesses that are likely generated by automated programs. We identify institutional IP addresses using the Chen et al. (2020) cipher together with ARIN and IAPD data. The SEC’s EDGAR log files record the IP address of each request only through June 2017. No logs are available from July 1, 2017 through May 18, 2020. The logs that the SEC releases from May 19, 2020 onward record only the time and path of each request and omit the IP addresses that we need to remove automated accesses and to identify institutional

¹⁰ Expected volume and its standard deviation are computed from *CRSP* daily volume over the 365 calendar days before the filing date and scaled to a 30-minute interval. *CRSP* daily volume includes trades that the previously described (Section 3.1) *TAQ* filters remove, so the standardized measure is negative on average. Our tests use only its variation across filings.

¹¹ QID^{res} exploits the fact that liquidity providers undercut one another less when they expect informed order flow. We compute it over the 30-minute pre-acceptance interval. We thank Barardehi, Dixon, and Liu for making their code and data publicly accessible. Coverage is concentrated in 2010Q2–2020Q1.

users. We therefore construct the view variables only through June 2017. Specifications containing these controls use the corresponding shorter sample period.

To capture return continuation at lower frequencies, we include six lagged returns. Return – overnight is the return from the previous trading day’s close to the starting price P_{-30} of the pre-acceptance window, adjusted for splits and distributions. Return – 1 day is the *CRSP* return on the previous trading day. Return – 7 days, Return – 30 days, Return – 180 days, and Return – 365 days are cumulative *CRSP* daily returns over the trading days in the corresponding number of calendar days before the filing date.

Table A.2 in the Appendix reports descriptive statistics for the baseline EXCL sample, Table A.3 reports its form-type composition, and Table A.5 reports the distributions over the years.

3.3. Filings with Significant Information Content

The leakage interpretation requires an incentive to acquire and trade on information before its release. That incentive exists only when the forthcoming filing has significant information content, because a filing that confirms prevailing expectations offers little to trade on. Filings carry significant information mainly when the firm has experienced major updates to its fundamentals, such that there are shifts in the operating cash-flow that the firm generates. We therefore select filings made when the firm’s operating cash-flow deviate significantly from what would have been predicted by recent firm and industry history.

We measure these deviations with a forecast of operating cash-flow (we use the phrases operating cash-flow, and cash-flow, interchangeably) that does not rely on analyst forecasts, because analyst forecasts can be biased and reported earnings can be strategically managed (e.g., [Graham, Harvey, and Rajgopal \(2005\)](#), [So \(2013\)](#), among others). We forecast quarterly operating cash-flow scaled by total assets using four lags each of the firm’s cash-flow, cash-flow growth, and book leverage and the corresponding leave-one-out NAICS-3 industry cash-flow measures. The model is estimated each year using the preceding six calendar years. It produces an out-of-sample R^2 of 61.7% relative to the test-set mean, and a benchmark-relative out-of-sample R^2 of 6.9% relative to the firm-specific four-quarter trailing mean. We define the magnitude of (operating) cash-flow news in a fiscal quarter as the absolute forecast

error, $|y - \hat{y}|$.

We then assign each filing to a firm-quarter. A firm-quarter runs from the day after the firm’s previous *IBES* quarterly earnings announcement through its next announcement, inclusive, and carries the cash-flow forecast error of the fiscal quarter reported at that announcement. The median firm-quarter spans 91 calendar days. We call a firm-quarter *informative* when its absolute forecast error exceeds the median across the 105,513 firm-quarters that contain at least one EXCL filing. Of the 157,154 EXCL filings with valid 30-minute returns, 149,439 fall in a firm-quarter with a cash-flow forecast error, and 68,569 fall in an informative firm-quarter. These 68,569 filings form the baseline EXCL sample. The same firm-quarter restriction applies to every form-type group, to the 90-minute windows, and to the placebo events of Section 5.1. We refer to this restriction as the cash-flow filter.

Because the cash-flow outcome of a quarter becomes public only at its earnings announcement, the filter conditions on the news of the period in which a filing is made rather than on information observable at acceptance. It selects the filings for which the leakage incentive is present, and the tests that follow ask whether prices move before acceptance within that set. Table A.1 in the Appendix shows that our main results remain statistically significant without the cash-flow filter.

3.4. The Unconditional Null Hypothesis

Our empirical null is no positive continuation across EDGAR acceptance. If pre-acceptance prices contain no filing-specific information, the pre-acceptance return should not positively predict the post-acceptance return. Moreover, unconditional intraday dynamics point in the opposite direction: [Heston, Korajczyk, and Sadka \(2010\)](#) show that half-hour stock returns tend to reverse. Zero continuation therefore provides a conservative benchmark for our tests.

We also estimate the corresponding relation at random firm–date–times that are unrelated to EDGAR events. These placebo observations provide a direct benchmark for half-hour return dynamics in the same set of stocks. Section 5.1 compares the filing and placebo estimates.

4. Main Results

4.1. Graphical Results

Figure 2 provides a model-free view of the baseline result. We sort filings into quintiles of the 30-minute pre-acceptance return and plot the mean cumulative log return for each quintile at 15-second intervals from 30 minutes before to 30 minutes after acceptance, where each 15-second return is the log change in the mean transaction price between consecutive intervals. When an interval contains no trade, the most recent earlier mean price is carried forward.

The quintiles continue to separate after SEC acceptance. Filings with the largest pre-acceptance gains continue to rise, while those with the largest losses continue to fall. This pattern contrasts with the unconditional half-hour reversal documented by [Heston, Korajczyk, and Sadka \(2010\)](#) and is consistent with the notion that pre-acceptance price movement contains information about the market’s subsequent response to the filing. Section 4.2 quantifies the continuation. If we do not use the baseline EXCL form-type group, Figure IA.1 still shows a similar pattern in the all-form-type sample, which includes Forms 3, 4, 5, SC 13D, and SC 13G.

4.2. Single Sort

Table 1 quantifies the pattern in Figure 2. Mean post-acceptance returns for the 30-minute interval (i.e., Q_{30}^{post}) decline monotonically from 4.42 basis points ($t = 4.84$) in the highest pre-acceptance-return quintile to -4.44 basis points ($t = -4.75$) in the lowest. The resulting top-minus-bottom spread in Q_{30}^{post} is 8.86 basis points ($t = 6.78$). To put these magnitudes in context, filings in the highest pre-acceptance-return quintile rise by an average of 92.41 basis points before acceptance and by a further 4.42 basis points after acceptance (in the 30-minute interval). Filings in the lowest quintile fall by an average of 94.14 basis points before acceptance and by a further 4.44 basis points after acceptance. Thus, the continuation is visible both in the extreme-quintile spread and across the full distribution of pre-acceptance returns.

The continuation appears across filing categories. The top-minus-bottom spread is 4.18

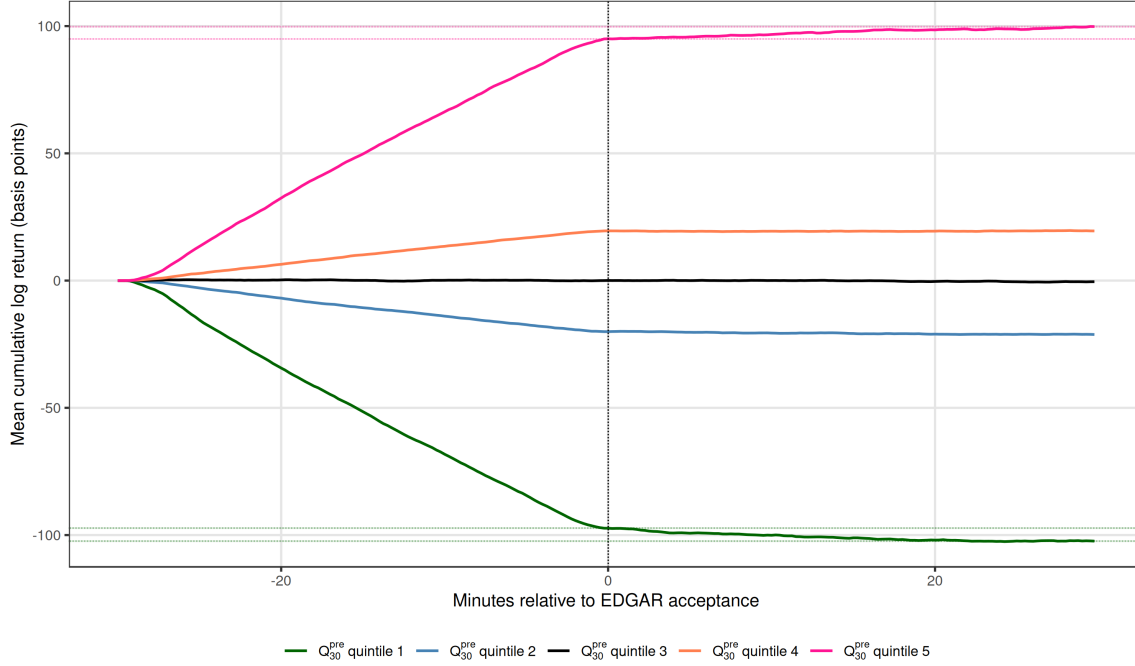


Figure 2: Cumulative returns around EDGAR acceptance by pre-acceptance-return quintile, EXCL form types

Note: The sample (EXCL form types) includes all form types excluding 3/4/5/13D/13G accepted by EDGAR between September 2003 and December 2024. Filings are restricted to informative firm-quarters (Section 3.3). Filings are sorted into quintiles of the 30-minute pre-acceptance return Q_{30}^{pre} . For each quintile, the figure plots the mean cumulative log return, in basis points, at 15-second intervals from 30 minutes before to 30 minutes after the EDGAR acceptance timestamp. When a 15-second interval has no trade, the most recent earlier mean price is carried forward. The vertical line marks the acceptance timestamp. The dotted horizontal lines mark the levels of quintiles 1 and 5 at acceptance and 30 minutes after acceptance.

basis points ($t = 5.22$) in the all-form-type sample (Table A.6), 5.68 basis points ($t = 2.44$) among the *scheduled* EXCL filings (Table A.7), 9.80 basis points ($t = 6.30$) among the *unscheduled* EXCL filings (Table A.8), 15.52 basis points ($t = 5.49$) for Form 8-K (Table A.9), and 7.52 basis points ($t = 2.20$) for Form 10-Q (Table A.10). For Form 10-K, the spread is 3.21 basis points ($t = 0.60$) across 2,841 filings (Table A.11).¹² Even after excluding Forms 8-K, 10-K, and 10-Q, the spread remains 3.71 basis points ($t = 2.63$; Table A.12). The

¹² For Form 10-K filings, in a panel regression of the post-acceptance return on the pre-acceptance return with firm and month fixed effects and standard errors clustered two-way by firm and calendar month, the regression coefficient is 0.1036 ($t = 1.83$).

pre-announcement drift is therefore not confined to a single filing category, in contrast to the single-form settings of prior studies.

The baseline t -statistic treats filings as independent observations, so we also account for dependence across firms and calendar months. Table [IA.2](#) reports the same top-minus-bottom spread with standard errors clustered by calendar month and clustered two-way by firm and month. Its t -statistic is 6.78 under independence, 6.60 with month clustering, and 6.71 with two-way clustering by firm and month. A month-block bootstrap yields a 95% confidence interval of [6.19, 11.57] basis points, and a within-month randomization test gives $p = 0.001$. Cross-sectional and calendar-time dependence therefore do not account for the baseline significance.

Common market movements also do not explain the result. Subtracting the contemporaneous S&P 500 return from both event returns reduces the top-minus-bottom spread only modestly, from 8.86 ($t = 6.78$) to 7.39 basis points ($t = 5.93$; Table [A.18](#)). In addition, the random-time placebo sample is exposed to the same aggregate market conditions but produces a spread of only 0.72 basis points, statistically indistinguishable from zero (Table [A.17](#)). The continuation therefore appears to be filing-specific rather than a manifestation of common intraday market movements.

4.3. Portfolio Difference

Table [IA.3](#) compares every pair of pre-acceptance-return quintiles. All ten post-acceptance-return differences have the predicted sign, and seven are significant at the five percent level. The exceptions are the three differences among the middle quintiles 2, 3, and 4. All ten pairwise differences are also positive in the all-form-type sample, among scheduled and unscheduled filings, and for the EXCL form types other than Forms 8-K, 10-K, and 10-Q (Tables [IA.4](#), [IA.5](#), [IA.6](#), and [IA.10](#)). For Forms 8-K, 10-Q, and 10-K, nine of the ten differences are positive (Tables [IA.7](#), [IA.8](#), and [IA.9](#)). The baseline result therefore does not depend only on the two extreme quintiles.

Q_{30}^{pre}	bin	Mean Q_{30}^{post} (bps)	t	Std. err. (bps)
	5	4.42***	4.84	0.91
	4	0.07	0.15	0.50
	3	-0.46	-1.13	0.40
	2	-1.07**	-2.26	0.47
	1	-4.44***	-4.75	0.93
	5–1	8.86***	6.78	1.31
Total Observations N = 68,569				

Table 1: Single sort, EXCL, 30-minute windows

Note: The sample (EXCL form types) includes all form types excluding 3/4/5/13D/13G, accepted by EDGAR between September 2003 and December 2024. Filings are restricted to informative firm-quarters (Section 3.3). Pre- and post-acceptance returns are measured over 30-minute windows around the EDGAR acceptance timestamp of each filing. Quintile means report one-sample t -statistics, and the 5–1 reports the Welch two-sample statistic. To indicate significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

4.4. Alternative Explanations

We next ask whether familiar sources of return continuation—investor attention, longer-horizon momentum, or common market movements—account for the pre-announcement drift. We estimate panel regressions of the post-acceptance return on the pre-acceptance return and the controls defined in Section 3.2, including firm and month fixed effects. Firm fixed effects absorb persistent firm characteristics, while month fixed effects absorb common time-series variation.

Table 2 reports the estimates with *standardized* regressors. The coefficient on the pre-acceptance return remains between 0.0658 and 0.0711 across specifications and is statistically significant throughout. In column (1), a one-standard-deviation increase in the pre-acceptance return, about 66 basis points (Table A.2), predicts a post-acceptance return that is about 4.4 basis points higher. Adding firm size, lagged returns, and three measures of EDGAR attention changes the coefficient little. The controls therefore do not absorb the predictive relation between pre- and post-acceptance returns.

The lagged-return coefficients also locate the predictive information in the half hour before acceptance. Adding the overnight, one-day, seven-day, and longer lagged returns moves the coefficient on the pre-acceptance return only from 0.0661 to 0.0658. Particularly telling,

is the definition of the overnight return control, recall again from page 13 that it is the return from the previous trading day’s close to the starting price P_{-30} of the pre-acceptance window. Notably, among these controls, only the previous day’s return is significant at the five percent level, and its coefficient of -0.0242 indicates reversal rather than continuation. The pre-announcement drift therefore reflects price movements immediately before acceptance rather than the firm’s return over the preceding day or week.

Month fixed effects absorb common variation at a monthly horizon, but filings accepted on the same day can also share market and disclosure conditions. The results are robust to fixed-effect and clustering specifications that absorb this daily variation. Table A.13 in the Appendix replaces the month fixed effects with filing-date fixed effects and clusters standard errors two-way by firm and filing date. The coefficient on the pre-acceptance return is between 0.0521 and 0.0568, with t -statistics between 3.36 and 3.82. Table A.14 uses filing-date fixed effects alone and clusters standard errors by filing date. The coefficient is between 0.0514 and 0.0560, with t -statistics between 3.29 and 3.62. The pre-announcement drift is significant at the one percent level in every specification of both tables. The continuation therefore does not depend on how the panel absorbs common time variation.

The same panel specifications produce no corresponding relation in the random-time placebo sample. As Table IA.11 shows, the pre-event return coefficient is statistically insignificant in every specification. The return continuation therefore appears specifically around filing acceptance rather than at arbitrary firm–date–times.

The result is also robust to removing contemporaneous market movements. When both event returns are measured relative to the S&P 500, the pre-acceptance coefficient remains positive and statistically significant across the specifications in Table A.15.¹³

4.4.1. Panel Regression after Controlling for Firm-Specific News Events

Finally, we test whether the pre-acceptance return reflects delayed adjustment to earlier firm-specific news.¹⁴ Following Jiang, Li, and Wang (2021), we identify firm-specific news events as *RavenPack* business stories in their twelve fundamental news groups that have a

¹³ We thank Neil Pearson for this suggestion.

¹⁴ We are indebted to Sophia Zhengzi Li for suggesting this exercise.

	Post-acceptance return				
	No control (1)	Size (2)	Returns (3)	Views (4)	All (5)
Pre-acceptance return	0.0661*** (5.04)	0.0661*** (5.03)	0.0658*** (5.01)	0.0711*** (4.47)	0.0707*** (4.42)
Log firm size		0.0108 (0.37)			0.0285 (0.58)
Return - overnight			0.0020 (0.19)		-0.0001 (-0.01)
Return - 1 day			-0.0242*** (-3.18)		-0.0257*** (-2.81)
Return - 7 days			-0.0064 (-0.66)		-0.0052 (-0.42)
Return - 30 days			-0.0003 (-0.03)		-0.0039 (-0.32)
Return - 180 days			-0.0068 (-0.91)		-0.0086 (-1.09)
Return - 365 days			-0.0085 (-0.76)		-0.0193 (-1.54)
Total views, 30 days before				0.0008 (0.56)	0.0015 (1.03)
Institutional views dummy, 30 days before				0.0057 (1.07)	0.0056 (1.03)
Institutional views, 30 days before				-0.0038 (-0.74)	-0.0041 (-0.80)
Observations	68,055	68,055	68,054	49,462	49,461
R ²	0.0752	0.0752	0.0762	0.0814	0.0826
Firm fixed effects	✓	✓	✓	✓	✓
Month fixed effects	✓	✓	✓	✓	✓
Two-way clustered <i>t</i> -statistics	✓	✓	✓	✓	✓

Table 2: Fixed-effect regression of the post-acceptance return on the pre-acceptance return, EXCL form types

Note: The table reports fixed-effect regressions of the 30-minute post-acceptance return on the 30-minute pre-acceptance return and the control variables defined in Section 3.2, with firm and month fixed effects. The sample is EXCL filings accepted by EDGAR between September 2003 and December 2024 in informative firm-quarters (Section 3.3). All variables are standardized, and standard errors are clustered two-way by firm and calendar month. Filings that are the only observation of their firm or of their calendar month drop from the estimation. *t*-statistics are displayed below coefficient estimates. To indicate significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

relevance score of 100 for the firm and no similar story about the firm in the preceding 365 days.¹⁵

We then remove every filing preceded by such a news event in the six calendar days before the filing date and re-estimate the panel regressions. Because *RavenPack* coverage is denser in later years, this restriction shifts the sample toward earlier observations; we therefore interpret it as a robustness subsample rather than a directly comparable specification. As Panel A of Table A.16 shows, the coefficient on the pre-acceptance return remains positive and statistically significant throughout.

News released earlier on the filing date could also reach prices before acceptance, for example through a press release whose content the filing later furnishes. We therefore also remove filings with a firm-specific news event between midnight and the acceptance timestamp on the filing date. Such news precedes 25.3% of the baseline filings. Among the remaining 51,193 filings, the top-minus-bottom spread is 8.68 basis points ($t = 5.62$), essentially identical to the baseline spread of 8.86 basis points. Removing news both in the six preceding days and earlier on the filing date leaves 35,942 filings, with a spread of 9.72 basis points ($t = 4.90$). Panel B of Table A.16 shows that the coefficient on the pre-acceptance return in this sample is positive and significant at the one percent level in every specification, between 0.0580 and 0.0622. Firm-specific news before acceptance, whether in the preceding days or earlier on the filing date, therefore does not account for the baseline relation.

5. Robustness Tests

5.1. Placebo Tests

To distinguish filing-specific return continuation from generic half-hour return dynamics, we construct placebo events at random firm–date–times. We draw *CRSP* permnos with replacement, select a date uniformly from each firm’s eligible *TAQ* trading days, and assign a clock time uniformly between 10:00 a.m. and 3:30 p.m. ET. We apply the same overlap restriction used for actual filings, and we retain placebo events that fall in informative

¹⁵ The 365-day event-similarity filter removes repeated coverage of an event and keeps the first report of each new event.

firm-quarters, so that the placebo sample shares the cash-flow news of the filing sample. This leaves 98,374 placebo events. By construction, these events are unrelated to EDGAR activity.

Table A.17 compares actual filings with placebo events. The top-minus-bottom spread over the 30-minute window is 8.86 basis points ($t = 6.78$) for genuine filings, and 0.72 basis points ($t = 0.59$) for the random-draw placebo events, producing a filing-minus-placebo difference of 8.14 basis points ($t = 4.57$). In pooled regressions with two-way clustering by firm and event date, the continuation slope is 0.0652 ($t = 4.90$) for filings and -0.0189 ($t = -1.11$) for placebo events; their difference is 0.0841 ($t = 3.90$). The baseline panels cluster two-way by firm and calendar month, the horizon over which filings share market and disclosure conditions. This slope instead clusters by event date, because the filing and placebo samples are compared on the same dates. The return continuation phenomenon is therefore established for genuine filing timestamps, but not for the placebo sample with randomly drawn timestamps.

A small fraction of placebo draws coincides with genuine filing activity: 3.33% fall on a day when the firm files during trading hours, 1.18% fall within 60 minutes, and only 0.62% fall within 30 minutes. Removing same-day, adjacent-day, or nearby filing observations leaves the placebo spread statistically indistinguishable from zero. Table IA.12 reports these checks. Accidental overlap with genuine EDGAR events does not account for the placebo results.

5.2. Alternative Event Price

We next recompute the results using the alternative event-price definition from Section 3.2. Starting the post-acceptance return at $[-3, 0)$ ensures that it captures the entire initial market reaction to the filing. This alternative event-price definition also increases the sample size, as it no longer requires valid actual trades over $[0, 3)$. The EXCL top-minus-bottom spread is 11.84 basis points ($t = 8.69$) across 72,381 filings (Table A.18), compared with 8.86 basis points across 68,569 filings under the baseline definition.

However, because the pre- and post-acceptance returns share the mean price over $[-3, 0)$, the alternative definition induces a mechanical positive correlation between the two returns. A price change during that interval enters the mean price only in proportion to the part of the interval that follows it, so the pre-acceptance return captures that part of the change

and the post-acceptance return captures the remainder, with the same sign.¹⁶ The filing-minus-placebo difference accounts for this mechanical correlation and nets it out. Table A.18 shows that the filing-minus-placebo difference remains between 7.46 and 8.14 basis points and is significant at the one percent level under both event-price definitions and for both raw and market-adjusted returns. Table IA.13 confirms that the sort, panel, and trading-activity results remain statistically significant under the alternative definition.

5.3. Quote-Midpoint Prices

We next replace transaction prices with NBBO midpoints, which do not require a trade in each three-minute endpoint interval. Using NBBO midpoints, the top-minus-bottom spread is 5.87 basis points ($t = 5.64$) for filings and essentially zero for placebo events (0.09 basis points, $t = 0.12$), yielding a filing-minus-placebo difference of 5.77 basis points ($t = 4.36$; Table IA.14). The midpoint sample is also substantially larger, 115,455 versus 68,569 filings, because a valid quote does not require a contemporaneous transaction. The continuation therefore survives both the removal of bid-ask bounce and the inclusion of less actively traded observations.

5.4. After Known Dissemination Defects and Hacking Episodes

Table IA.15 restricts the 30-minute EXCL sample to filings on or after 1 March 2015, after the SEC fixed the dissemination-system defect documented by Bolandnazar et al. (2020). Table IA.16 restricts the sample to filings on or after 1 October 2020, after the alleged filing-agent hacking window. The top-minus-bottom spread is 5.58 basis points ($t = 2.85$) after March 2015; and 7.49 basis points ($t = 3.19$) after October 2020, which is notable in that the magnitude and statistical significance both increased for the post October 2020 sample (compared to the post March 2015 sample), despite the smaller sample size due to shorter sample period. Both estimates are significant at the one percent level.

¹⁶ For example, if the price changes from \$100.00 to \$101.00 at $t = -2$ and remains constant otherwise, then the mean price over $[-3, 0)$ is \$100.67. Even if prices never change outside of $[-3, 0)$, pre- and post-acceptance returns would both be positive.

5.5. Pre-Event Returns over Longer Horizons

We next ask whether the pre-announcement drift appears when the signal is measured over a longer horizon before the filing. Table 3 replaces the 30-minute pre-acceptance return with the firm’s return on the previous trading day or over the previous seven calendar days as the sorting variable, holding the sample and the post-acceptance return fixed. Sorting on the 30-minute pre-acceptance return yields the baseline spread of 8.86 basis points ($t = 6.78$). Sorting on the previous day’s return instead yields a spread of -4.85 basis points ($t = -3.95$), and sorting on the seven-day return yields a spread of -3.43 basis points ($t = -2.88$). The longer-horizon returns therefore predict reversal rather than continuation.

The pre-announcement drift is thus concentrated in the half hour before the acceptance timestamp. A longer pre-event horizon accumulates price movements unrelated to the forthcoming filing, so its signal-to-noise ratio for filing information is lower, and price-based detection of leakage over such a horizon becomes difficult. Over the previous day or week, the reversal of transient return noise overwhelms any continuation that information leakage produces.

5.6. Additional Robustness Tests

Two additional tests appear in the Internet Appendix. First, following [Bargeron, Clifford, and Qiu \(2025\)](#), we classify observations as self-filed or agent-filed from accession numbers; pre-announcement drift appears in both groups (Table IA.17). The top-minus-bottom spread is 11.76 basis points ($t = 4.45$) across 14,203 self-filed forms, which is larger in magnitude than that of 7.89 basis points ($t = 5.28$) across 54,366 agent-filed forms. The difference between the two groups is not statistically significant ($t = 1.54$). Because self-filed forms bypass filing agents, the significant drift among self-filed forms cannot arise from the filing-agent channel. Our sample differs from theirs: [Bargeron, Clifford, and Qiu \(2025\)](#) study Form 8-K releases after trading hours, whereas we study a broad set of filings accepted during trading hours. Within our sample, the filing method (agent-filing versus self-filing) does not explain the pre-announcement drift.

Second, we study the impact of extending the event time window on the results. Extending

Sorting variable	Quintile 5 (bps)	Quintile 1 (bps)	5–1 (bps)	<i>t</i>
Q_{30}^{pre}	4.42***	-4.44***	8.86***	6.78
Return – 1 day	-2.57***	2.29***	-4.85***	-3.95
Return – 7 days	-2.67***	0.76	-3.43***	-2.88
Total Observations N = 68,568				

Table 3: Pre-announcement drift by pre-event return horizon

Note: The sample (EXCL form types) includes all form types excluding 3/4/5/13D/13G accepted by EDGAR between September 2003 and December 2024. Filings are restricted to informative firm-quarters (Section 3.3). Each row sorts filings into quintiles of the named pre-event return and reports the mean 30-minute post-acceptance return in the top and bottom quintiles and their difference. Q_{30}^{pre} is the 30-minute pre-acceptance return. Return – 1 day is the *CRSP* return on the previous trading day, and Return – 7 days is the cumulative *CRSP* daily return over the trading days in the seven calendar days before the filing date, as defined in Section 3.2. All rows use the filings for which every sorting variable is available. Quintile means report one-sample *t*-statistics, and the 5–1 reports the Welch two-sample statistic. To indicate significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

both pre- and post-event windows from 30 to 90 minutes produces a top-minus-bottom spread of 10.75 basis points ($t = 3.76$; Table IA.18).

6. Evidence on the Leakage Interpretation

If the pre-announcement drift reflects filing information reaching some investors before public dissemination, the leakage interpretation yields two additional predictions. First, large pre-acceptance price moves should coincide with intense, information-sensitive (i.e., carrying an elevated informed-trading risk from the perspective of market makers), and directional trading before acceptance. Second, leakage may be more prevalent when regulatory monitoring and enforcement capacity are more constrained. Section 6.1 examines pre-event trading activity, and Section 6.2 examines the SEC’s enforcement caseload.

6.1. Pre-Event Trading Activity

To corroborate the price-based evidence, we examine the trading-activity measures defined in Section 3.2. If the pre-acceptance return contains filing-related private information, large price moves before acceptance should *coincide* with unusually intense trading or with trading

that carries high informed-trading risk from the perspective of market makers. Figure 3 confirms this pattern. Both abnormal volume and the informed trading risk measure— QID^{res} (Barardehi, Dixon, and Liu (2026)) are U-shaped across the pre-acceptance-return quintiles, with the highest values in the two extremes. In other words, large pre-acceptance price moves in either direction coincide with high abnormal volume and high QID^{res} . Recall from previous discussions that filings in the two extreme pre-acceptance-return quintiles also have the largest subsequent post-acceptance returns in absolute value.

We next ask whether continuation is stronger when pre-event trading activity is more informative. Table 4 sorts filings independently into quintiles of the pre-acceptance return and into quintiles of abnormal volume or QID^{res} . For abnormal volume, the top-minus-bottom return spread rises from 0.47 basis points in the lowest-volume bin to 17.42 basis points in the highest; the difference is 16.95 basis points ($t = 4.47$). For QID^{res} , the corresponding spread rises from 2.94 to 17.21 basis points, a difference of 14.27 basis points ($t = 2.24$). Continuation is therefore concentrated in filings with unusually high pre-event trading activity and informed-trading risk.

Directional order flow provides complementary evidence. Table A.19 shows that the Lee–Ready imbalance increases monotonically with the pre-acceptance return: seller-initiated volume dominates in the bottom quintile, while buyer-initiated volume dominates in the top. The extreme-quintile difference in X_{LR} is 0.2458 across 68,563 filings. X_{LR} is unavailable for 6 of the 68,569 baseline filings, because none of their pre-acceptance trades can be signed. In our baseline analysis, we require trades in both pre- and post-acceptance endpoint intervals. Because X_{LR} requires only pre-acceptance trades, we also relax the requirement that post-acceptance returns be available. This adds 9,262 filings and results in an extreme-quintile difference of 0.2680.

Together, these results show that the pre-acceptance return reflects executed directional order flow rather than only quote noise or bid–ask bounce. Abnormal volume and QID^{res} further indicate that this order flow is more consistent with information-based trading than with purely mechanical trade direction.

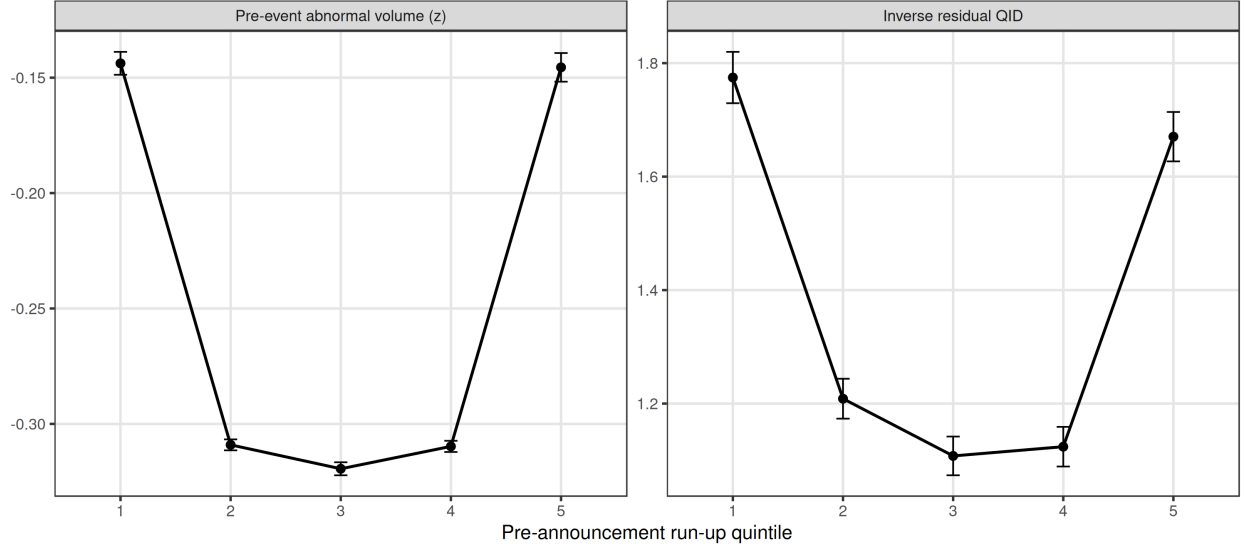


Figure 3: Pre-event abnormal volume and inverse residual QID (QID^{res}) by pre-acceptance-return quintile, EXCL form types, September 2003 to December 2024

Note: The sample (EXCL form types) includes all form types excluding 3/4/5/13D/13G accepted by EDGAR between September 2003 and December 2024. Filings are restricted to informative firm-quarters (Section 3.3). Pre-acceptance returns are measured over the 30 minutes before the EDGAR acceptance timestamp of each filing. Quintiles are formed on Q_{30}^{pre} . The left panel is mean 30-minute pre-event abnormal volume. The right panel is mean 30-minute inverse residual QID (i.e., QID^{res}). QID^{res} data is available from 2010Q2 to 2020Q1. Error bars are plus or minus one standard error.

6.2. SEC Enforcement Caseload

We examine the second prediction by asking whether pre-announcement drift covaries with the pressure on the SEC’s enforcement resources.¹⁷ We measure enforcement workload using the NYU Pollack Center for Law & Business/Cornerstone Research *Securities Enforcement Empirical Database* (*SEED*), which covers SEC actions against public companies and their subsidiaries from October 2009 through September 2023. For each month, *SEED* caseload is the average daily number of open actions during the preceding calendar month. An action is open from the date it is filed until the latest final resolution among its defendants. Actions without a final resolution remain open through the end of *SEED* coverage in September 2023.

¹⁷ Per Fulmer, Knill, and Yu (2023): “resource limitations constrain the activities of government agencies and affect the overall effectiveness of their enforcement.”

<i>Panel A. Abnormal volume</i>						
Q_{30}^{pre}	Abnormal volume bin					5–1
	5	4	3	2	1	
5	9.47*** (3.97)	4.88*** (2.79)	3.23** (2.15)	-1.33 (-0.84)	0.69 (0.57)	
4	-0.77 (-0.31)	0.50 (0.37)	1.14 (1.20)	-0.07 (-0.09)	-0.47 (-0.97)	
3	-4.41** (-2.49)	-0.28 (-0.25)	0.79 (0.86)	0.56 (0.77)	-0.37 (-0.86)	
2	-2.20 (-1.07)	-0.48 (-0.36)	-0.74 (-0.77)	-1.01 (-1.37)	-1.21** (-2.11)	
1	-7.95*** (-3.51)	-3.92** (-2.03)	-3.81** (-2.11)	-2.45* (-1.68)	0.22 (0.15)	
5–1	17.42*** (5.29)	8.80*** (3.38)	7.04*** (3.00)	1.12 (0.52)	0.47 (0.25)	16.95*** (4.47)
<i>Panel B. Inverse residual QID</i>						
Q_{30}^{pre}	Inverse residual QID bin					5–1
	5	4	3	2	1	
5	15.76*** (4.03)	3.21 (1.08)	1.61 (0.59)	-1.43 (-0.50)	3.61 (1.24)	
4	2.38 (1.47)	-1.02 (-0.80)	0.86 (0.68)	0.04 (0.03)	-1.55 (-1.41)	
3	2.07 (1.53)	0.26 (0.23)	1.01 (1.02)	0.27 (0.28)	0.80 (0.60)	
2	-1.03 (-0.58)	-2.84 (-1.36)	-1.27 (-1.06)	-1.31 (-1.13)	0.15 (0.13)	
1	-1.45 (-0.48)	-5.11 (-1.31)	-1.16 (-0.41)	-3.25 (-1.04)	0.67 (0.24)	
5–1	17.21*** (3.47)	8.32* (1.70)	2.77 (0.70)	1.82 (0.43)	2.94 (0.74)	14.27** (2.24)
Total Observations N = 68,569 (volume); 25,300 (QID^{res})						

Table 4: Post-acceptance return by pre-acceptance-return quintile and pre-event trading intensity

Note: The sample (EXCL form types) includes all form types excluding 3/4/5/13D/13G accepted by EDGAR between September 2003 and December 2024. Filings are restricted to informative firm-quarters (Section 3.3). Pre- and post-acceptance returns are measured over 30-minute windows around the EDGAR acceptance timestamp of each filing. Rows are quintiles of the pre-acceptance return, and columns are bins of the pre-event characteristic, with 5 denoting the highest value in each dimension. The two sorts are independent: pre-acceptance-return breakpoints are computed over the full EXCL estimation sample, and characteristic breakpoints over all filings with a non-missing characteristic. QID^{res} is available from 2010Q2 to 2020Q1. Interior cells report mean post-acceptance returns. The bottom row reports the return spread between pre-acceptance-return quintiles 5 and 1 within each characteristic bin. The bottom-right cell reports the difference between this spread in characteristic bins 5 and 1. Interior cells report one-sample t -statistics, the 5–1s report Welch two-sample statistics, and the corner cell tests the difference of the two extreme-bin 5–1s. To indicate significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 5 interacts the pre-acceptance return with SEC caseload in a panel regression with firm and month fixed effects. The regression excludes filings made while the filing firm itself is a defendant in an open *SEED* action, so that the caseload reflects pressure on SEC resources rather than scrutiny of the filing firm itself. The main effect of caseload is absorbed

by the month fixed effects. At the mean caseload, the coefficient on the pre-acceptance return is 0.0613 ($t = 3.57$), and a one-standard-deviation increase in SEC caseload raises it by 0.0560 ($t = 3.18$), nearly doubling the pre-announcement drift. To assess whether firm characteristics account for this relation, column (2) adds the full set of control variables from Table 2. Because the availability of the view variables ends in June 2017 (Section 3.2), this specification uses the 23,006 filings accepted between November 2009 and June 2017. Despite the smaller sample, the pre-announcement drift at the mean caseload remains significant, with a coefficient of 0.0661 ($t = 2.57$). The interaction with SEC caseload also remains significant, with a coefficient of 0.0730 ($t = 3.02$). Thus, pre-announcement drift is stronger when the SEC enforcement workload is high. These time-series associations are consistent with an enforcement channel, although they do not by themselves identify a causal effect of SEC resources on information leakage.

7. Economic Significance

Having established the pre-announcement drift and the evidence that links it to leakage, we assess its economic significance in two ways: the implied profits of informed trading before acceptance, and the returns to a simple intraday trading rule that trades after acceptance.¹⁸

7.1. Implied Profits from Pre-Filing Informed Trading

We first estimate bounds on the profits that could be associated with informed trading before filing acceptance, adapting the approach of [Bernile, Hu, and Tang \(2016\)](#). The lower bound attributes only the observed Lee–Ready order imbalance to informed market orders and allows imbalances in the wrong direction to generate losses. The upper bound attributes every transaction in the pre-acceptance window to informed trading on the side indicated by the pre-acceptance return. Both calculations use only information available by acceptance and value the resulting position at the end of the post-acceptance window. We compute the bounds separately for 30- and 90-minute windows.

Table 6 reports the implied profit bounds by pre-acceptance-return quintile. In the

¹⁸ We thank Chuck Trzcinka for suggesting the economic-significance analysis.

	Post-acceptance return	
	(1)	(2)
Pre-acceptance return	0.0613*** (3.57)	0.0661** (2.57)
Pre-acceptance return $\times$ <i>SEED</i> open cases	0.0560*** (3.18)	0.0730*** (3.02)
Observations	37,775	23,006
R ²	0.0980	0.1188
Controls		✓
Firm fixed effects	✓	✓
Month fixed effects	✓	✓
Two-way clustered <i>t</i> -statistics	✓	✓

Table 5: Pre-announcement drift and SEC enforcement caseload

Note: The table reports panel regressions of the 30-minute post-acceptance return on the 30-minute pre-acceptance return and its interaction with SEC open cases, with firm and month fixed effects. SEC open cases is the average daily number of open actions in *SEED* in the calendar month before the filing; an action is open from its filing date to the latest final resolution among its defendants. The sample includes EXCL filings accepted by EDGAR between November 2009 and September 2023, excluding filings made while the filing firm is a defendant in an open *SEED* action. Filings are restricted to informative firm-quarters (Section 3.3). Pre- and post-acceptance returns are measured over 30-minute windows around the EDGAR acceptance timestamp of each filing. Column (2) adds the control variables of Table 2, whose coefficients are not reported, and includes filings accepted through 30 June 2017, when the IP-level EDGAR logs end. All variables are standardized. The main effect of SEC open cases is absorbed by the month fixed effects and is not reported. All standard errors are clustered two-way by firm and calendar month. Filings that are the only observation of their firm or of their calendar month drop from the estimation. *t*-statistics are displayed below coefficient estimates. To indicate significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

30-minute specification, the lower bound averages \$1,637 per filing across the two extreme quintiles, while the upper bound averages \$25,818. The corresponding 90-minute bounds are \$3,000 and \$83,215 per filing. Trading aligned with the pre-acceptance return would therefore have earned between roughly \$1,600 and \$26,000 per extreme-quintile filing over the 30-minute window. These estimates quantify the potential gains from informed trading under the stated assumptions; they do not necessarily independently establish that the observed order flow reflects private information.

	Mean profit per filing (\$), by pre-acceptance-return quintile					
	5	4	3	2	1	1 and 5
<i>Panel A: 30-minute windows</i>						
Lower bound	1,885	-317	-217	-73	1,398	1,637
Upper bound	25,133	3,744	1,196	5,825	26,503	25,818
<i>Panel B: 90-minute windows</i>						
Lower bound	3,444	-91	-1,526	-515	2,549	3,000
Upper bound	79,785	25,747	11,279	29,317	86,646	83,215

Table 6: Implied profits from pre-filing informed trading

Note: The table reports implied profits associated with pre-acceptance informed trading under the bounding assumptions of [Bernile, Hu, and Tang \(2016\)](#). The lower bound attributes only the Lee–Ready order imbalance to informed market orders; the upper bound attributes all pre-acceptance volume to informed trading in the direction of the pre-acceptance return. Positions are accumulated during the 30- or 90-minute pre-acceptance window and valued at the end of the corresponding post-acceptance window. Quintiles are formed separately within each window length. “1 and 5” reports the average across the two extreme quintiles. We apply this methodology to the whole pre-acceptance window and revise the lower bound downward to adjust for the timing of trades within the window. The adjustment is estimated from every baseline EXCL filing on 264 randomly drawn trading days. The sample is EXCL filings accepted by EDGAR between September 2003 and December 2024. Filings are restricted to informative firm-quarters (Section [3.3](#)).

7.2. A Post-Acceptance Trading Rule

To assess how much of the information in the pre-acceptance return remains unpriced at acceptance, we construct a trading rule that trades only after acceptance. The trading rule consists of five portfolio units and trades only on EXCL filings. The rule uses only information available at the time of the trade. It therefore trades on every EXCL filing regardless of the firm-quarter’s cash-flow forecast error, which becomes known only at the next earnings announcement. The rule would still trade on the filings that the 30-minute overlap restriction of Section [3.1](#) would have removed, because a trader in real time can not be sure about whether the current filing would have been inactivated by a nearby filing by the same firm under the no overlap restriction. The long and short cutoffs are the previous calendar year’s 80th and 20th percentiles of the pre-acceptance return, respectively, computed on these filings. Each position is entered three minutes after the acceptance timestamp and held for 27 minutes. To deal with the potential of multiple filings by the same firm with overlapping 30-minute intervals, a portfolio unit takes a new signal only once its current position is closed. We mark positions to the quote midpoint for our baseline results, and use

the last transaction price (instead of quote midpoint) as a robustness check. Because we calculate 2004 quintile cutoffs based on 2003 filings, the trading rule runs from January 2004 through December 2024. A unit’s daily return compounds the returns on its trades that day, and an idle unit earns zero. The daily return of the rule is the equal-weighted average across the five units, and its excess return subtracts the daily risk-free rate. Figure 4 plots the value of one dollar invested in the daily excess returns of the rule.

Table 7 reports factor-model alphas for the monthly returns generated by the rule. At quote midpoints, the five-factor alpha is 1.81% per month ($t = 3.47$) and the annualized Sharpe ratio is 1.33. Using the last transaction price yields a five-factor alpha of 1.62% ($t = 2.85$) and a Sharpe ratio of 1.10. Alphas are significant at the one percent level in every specification under both prices. Because these calculations do not model the full costs of intraday execution, we interpret them as measures of the information that remains unpriced at acceptance rather than as attainable net trading returns.¹⁹

8. Conclusion

We document pre-acceptance price discovery around trading-hour SEC filings from 2003 through 2024. Because leakage requires an incentive, which exists only when a filing carries significant information, we focus on filings made in firm-quarters with significant cash-flow news. Returns during the 30 minutes before EDGAR acceptance positively predict returns during the 30 minutes afterward: in our baseline sample, the top-minus-bottom spread is 8.86 basis points. The relation appears across filing categories and horizons, including both filings that firms file themselves and those filed through filing agents. It survives alternative price definitions, quote-midpoint returns, market adjustment, and filing-date fixed effects, persists after documented SEC dissemination defects and hacking episodes—in fact, the post October 2020 subsample results are stronger than the post March 2015 subsample results; moreover, the relation would be absent at random firm–date–times, that is, if genuine filing date-times are replaced by placebo with random date-times. Our main empirical findings

¹⁹ Standard errors are Newey–West with 12 lags. Following other intraday studies that measure return predictability at quote midpoints (e.g., [Heston, Korajczyk, and Sadka \(2010\)](#), [Lou, Polk, and Skouras \(2019\)](#), [Bogousslavsky \(2021\)](#)), we treat the midpoint returns as a measure of information value rather than as the performance of a particular execution strategy.

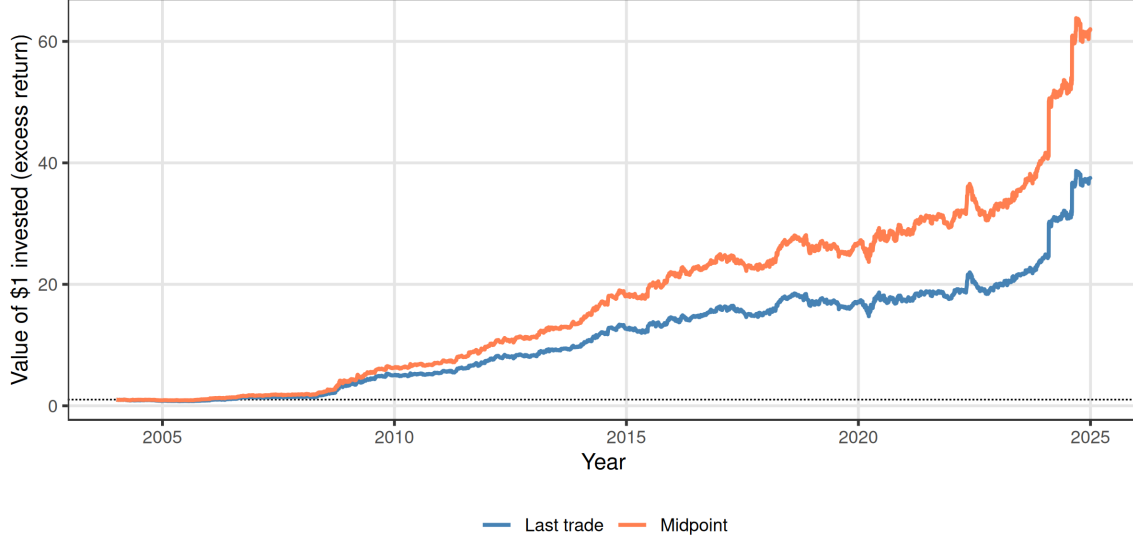


Figure 4: Value of \$1 invested in the five-unit trading rule, January 2004 to December 2024

Note: The figure plots the value of \$1 invested in January 2004 that compounds the daily excess returns of the trading rule. The horizontal line marks the initial value of \$1. The long and short cutoffs are the previous calendar year's 80th and 20th percentiles of Q_{30}^{pre} , respectively, computed on EXCL filings. The rule trades every EXCL filing, in all firm-quarters and including filings removed by the 30-minute overlap restriction. Each position enters three minutes after acceptance and is held for 27 minutes. A portfolio unit takes a new position only after its current position is closed. Last trade marks each position at the mean transaction price of the 15-second interval that begins three minutes after acceptance and of the interval that ends 30 minutes after acceptance. Midpoint marks each position at the mean NBBO midpoint of the same intervals. When an interval has no trade or quote update, the most recent earlier value is carried forward.

remain statistically significant without the cash-flow filter.

Various trend controls, EDGAR-based investor attention, and recent firm-specific news do not explain the relation. Sorting on the previous day's or week's return instead of the 30-minute pre-acceptance return produces reversal rather than continuation.

Several results connect this continuation to information about the forthcoming filing. Extreme pre-acceptance returns coincide with elevated abnormal volume, informed-trading risk measure (QID^{res}), and directional order flow in the pre-acceptance period. These three measures corroborate the model-free pre-acceptance return measure used in our baseline analysis. The pre-announcement drift is also strongest among filings with the highest pre-acceptance abnormal volume or informed-trading risk. The drift is also stronger when the

	FF3 (1)	FF3+Mom. (2)	FF5 (3)	FF5+Mom. (4)	JKP13 (5)
<i>Panel A: Quote midpoint</i>					
Alpha	0.0178*** (3.61)	0.0179*** (3.61)	0.0181*** (3.47)	0.0182*** (3.49)	0.0167*** (4.71)
R ²	0.0100	0.0103	0.0109	0.0112	0.0997
<i>Panel B: Last transaction price</i>					
Alpha	0.0161*** (2.93)	0.0161*** (2.94)	0.0162*** (2.85)	0.0163*** (2.87)	0.0141*** (3.79)
R ²	0.0108	0.0109	0.0112	0.0113	0.1030
Months	252	252	252	252	252
Newey–West, 12 lags	✓	✓	✓	✓	✓

Table 7: Monthly alphas of the five-unit trading rule, January 2004 to December 2024

Note: The dependent variable is the monthly excess return of the trading rule in Figure 4. The rule trades every EXCL filing, in all firm-quarters and including filings removed by the 30-minute overlap restriction. FF3 is the Fama–French three-factor model (Fama and French (1993)), FF5 adds profitability and investment (Fama and French (2015)), Mom. is the momentum factor (Carhart (1997)), and JKP13 is the 13 thematic factors of Jensen, Kelly, and Pedersen (2023). t -statistics, in parentheses, use Newey–West standard errors with 12 lags. To indicate significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

SEC’s enforcement caseload is high, consistent with the notion of leakage increasing when the SEC has less capacity to detect it.

The pre-announcement drift is economically meaningful. The implied profit of informed trading suggest economically significant incentives to trade ahead of filing acceptance. A trading rule that enters after acceptance and uses only information available at entry earns a monthly five-factor alpha of 1.81% at quote midpoints, and 1.62% at transaction prices, indicating that part of the information remains unpriced at acceptance. If we look at the JKP alphas (Jensen, Kelly, and Pedersen (2023)), the monthly alpha would be 1.67% at quote midpoints, and 1.41% at transaction prices.

Taken together, the evidence establishes pre-acceptance price discovery around SEC filings as a broad empirical phenomenon. The results are consistent with information from forthcoming filings reaching some market participants before public dissemination, although we do not directly observe the source or transmission channel. More broadly, the findings imply that formal public release need not mark the point at which corporate information first enters market prices, and they highlight the importance of institutions that protect

information before disclosure.

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Appendix

I. Results with and without the Operating Cash Flow Based Informativeness Filter

	Above-median cash-flow news		All firm-quarters	
	Estimate	N	Estimate	N
<i>Single sorts, 5–1 of Q_{30}^{post} (bps)</i>				
EXCL	8.86*** (6.78)	68,569	6.03*** (6.90)	157,154
All form types	4.18*** (5.22)	145,513	3.12*** (5.74)	321,842
Scheduled	5.68** (2.44)	17,072	3.95*** (2.73)	38,925
Unscheduled	9.80*** (6.30)	51,497	6.74*** (6.36)	118,229
Form 8-K	15.52*** (5.49)	23,658	11.53*** (5.56)	51,789
Form 10-K	3.21 (0.60)	2,841	0.04 (0.01)	5,943
Form 10-Q	7.52** (2.20)	9,930	6.85*** (3.23)	22,078
EXCL excluding Forms 8-K, 10-K, and 10-Q	3.71*** (2.63)	32,140	2.61*** (3.09)	77,344
90-minute window	10.75*** (3.76)	34,780	9.29*** (5.30)	79,961
Filings on or after 1 March 2015	5.58*** (2.85)	24,459	3.19** (2.45)	61,525
Filings on or after 1 October 2020	7.49*** (3.19)	11,142	4.82** (2.20)	26,063
<i>Placebo comparison</i>				
Placebo 5–1 (bps)	0.72 (0.59)		1.29 (1.45)	
Filing-minus-placebo 5–1 (bps)	8.14*** (4.57)		4.74*** (3.81)	
Filing-minus-placebo continuation slope	0.0841*** (3.90)		0.0772** (2.43)	
<i>Panel regressions, coefficient on Q_{30}^{pre} (standardized)</i>				
No controls	0.0661*** (5.04)	68,055	0.0657*** (2.95)	156,614
All controls	0.0707*** (4.42)	49,461	0.0607*** (2.66)	111,443
<i>Implied profits, extreme quintiles, 30-minute windows (\$ per filing)</i>				
Lower bound	1,637		1,630	
Upper bound	25,818		24,550	
<i>Pre-event trading activity</i>				
Abnormal volume double sort, 5–1 of 5–1 (bps)	16.95*** (4.47)		13.02*** (4.99)	
QID^{res} double sort, 5–1 of 5–1 (bps)	14.27** (2.24)		7.32** (2.16)	
Lee–Ready order imbalance, 5–1 of X_{LR}	0.2458*** (75.56)		0.2339*** (110.51)	
<i>SEC enforcement caseload, previous month, interaction with Q_{30}^{pre} (standardized)</i>				
SEED open cases	0.0560*** (3.18)	37,775	0.0477*** (3.39)	93,501

Table A.1: Main results with and without the operating cash flow based informativeness filter

Note: The first two columns report each statistic in the baseline sample of informative firm-quarters (with the above-median operating cash flow forecast deviation based informativeness filter applied as described in Section 3.3), and the last two columns report it on all firm-quarters without the aforementioned filter. The randomly-drawn placebo sample (as constructed in Section 5.1) of the first two columns is restricted to the same informative firm-quarters. Entries are point estimates with the t -statistic of the summarized exhibit in parentheses: Welch two-sample statistics for sorts and double sorts, two-way clustered statistics by firm and calendar month for the panel regression and SEC-enforcement rows, and two-way clustered statistics by firm and event date for the placebo slope. The implied-profit rows report mean dollar profits per extreme-quintile filing, and the SEC enforcement row reports the interaction coefficient in column (1) of Table 5. To indicate significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

II. Descriptive Statistics

Variable	N	Mean	SD	p10	p25	p50	p75	p90
<i>Pre- and post-acceptance returns</i>								
Q_{30}^{pre} (bps)	68,569	-0.39	66.03	-65.73	-25.56	0.00	24.67	64.15
Q_{30}^{post} (bps)	68,569	-0.20	60.91	-61.98	-24.36	0.00	23.84	61.06
Q_{30}^{pre} , market-adj. (bps)	68,546	-0.22	60.41	-57.61	-22.48	-0.29	21.74	57.42
Q_{30}^{post} , market-adj. (bps)	68,555	-0.24	54.95	-54.24	-21.29	-0.29	20.65	53.78
<i>Pre-event trading activity</i>								
Pre-event volume, 30 min (000 sh.)	68,569	98.33	204.53	4.70	10.30	27.27	83.48	244.09
Pre-event dollar volume (\$000)	68,569	4,129	8,108	81	241	888	3,760	12,049
Trades in pre-event window	68,569	1,143	1,777	93	188	465	1,254	2,972
Abnormal volume, 30 min (z)	68,569	-0.26	0.27	-0.54	-0.43	-0.29	-0.16	0.01
Inverse residual QID	25,300	1.35	2.53	-1.51	-0.31	1.08	2.75	4.64
Lee-Ready order imbalance X_{LR}	68,563	0.00	0.28	-0.33	-0.16	0.00	0.16	0.34
<i>Announcing firm characteristics</i>								
Market capitalization (\$m)	68,569	30,046	69,980	398	976	3,359	17,316	88,335
Log book-to-market	45,381	-0.92	0.91	-2.11	-1.45	-0.81	-0.26	0.15
<i>Investor attention</i>								
Total views, 30 days before	49,960	815	2,072	53	112	242	556	1,466
Institutional view dummy, 30 days before	49,960	0.54	0.50	0	0	1	1	1
Institutional views, 30 days before	49,960	4.72	10.72	0	0	1	4	13
<i>Price momentum</i>								
Return - overnight (%)	68,569	0.08	3.41	-3.06	-1.17	0.03	1.26	3.22
Return - 1 day (%)	68,568	0.07	3.02	-3.09	-1.24	0.04	1.36	3.17
Return - 7 days (%)	68,569	0.33	6.57	-6.81	-2.78	0.28	3.29	7.34
Return - 30 days (%)	68,569	1.53	13.14	-13.17	-5.22	1.31	7.80	16.08
Return - 180 days (%)	68,569	8.34	33.17	-28.59	-10.61	6.13	22.75	44.10
Return - 365 days (%)	68,569	18.33	54.11	-36.75	-13.32	10.52	37.59	75.84

Table A.2: Descriptive statistics, EXCL estimation sample

Note: The table reports filing-level statistics for the 68,569 EXCL filings in the 30-minute estimation sample, accepted by EDGAR between September 2003 and December 2024 in informative firm-quarters (Section 3.3). Event returns are in basis points. N varies across rows because some variables are available only for part of the sample: the view variables end on 30 June 2017, the last date of the IP-level EDGAR server-logs, book-to-market requires a *Compustat* match, and the inverse residual QID is concentrated in 2010Q2–2020Q1. Views over 30 days count views of all of the firm’s filings in the 30 calendar days before the filing date. Lagged returns are in percent. Means and standard deviations are winsorized at the 1st and 99th percentiles, except for variables whose 1st and 99th percentiles coincide; the reported percentiles are not winsorized.

Form	N	% of sample	Mean Q_{30}^{pre} (bps)	Mean Q_{30}^{post} (bps)
8-K	22,984	33.5	0.00	0.61
10-Q (with amendments)	9,930	14.5	-0.24	-0.60
424B2	5,932	8.7	0.11	-1.36
UPLOAD	4,245	6.2	-1.20	-0.24
10-K (with amendments)	2,841	4.1	1.29	0.13
144	2,661	3.9	-0.83	-1.91
CORRESP	2,330	3.4	-1.89	-2.52
ARS	2,033	3.0	-2.37	0.89
DEF 14A	1,680	2.5	-1.95	1.36
FWP	1,461	2.1	0.47	1.46
11-K	1,174	1.7	-2.84	-1.44
CT ORDER	1,087	1.6	-0.50	2.99
DEFA14A	975	1.4	-0.76	0.73
424B3	893	1.3	0.90	-1.63
S-8	878	1.3	-1.07	-2.50
425	799	1.2	-1.07	1.62
NO ACT	682	1.0	3.76	2.84
8-K/A	673	1.0	-1.69	3.02
SD	576	0.8	-0.50	0.20
424B5	529	0.8	-6.73	-2.70
All other form types (136)	4,206	6.1	-2.07	-4.09

Table A.3: Form-type composition of the EXCL estimation sample

Note: The table reports the 20 most frequent form types in the EXCL estimation sample. The remaining form types are pooled in the last row. The 10-K and 10-Q rows include their amendments and transition reports. Mean returns are in basis points.

Form type	Description
<i>Periodic reports of operating companies</i>	
10-K	Annual report of a domestic issuer
10-Q	Quarterly report of a domestic issuer
20-F	Annual report of a foreign private issuer
40-F	Annual report of a Canadian issuer under the multijurisdictional disclosure system
ARS	Annual report to security holders
11-K	Annual report of an employee stock purchase, savings, or similar plan
SD	Specialized disclosure report, chiefly on conflict minerals, due 31 May each year
<i>Asset-backed issuers</i>	
10-D	Periodic distribution report of an asset-backed issuer
<i>Registered investment companies</i>	
N-CSR	Certified annual report to shareholders
N-CSRS	Certified semiannual report to shareholders
N-30D	Annual or semiannual report to shareholders, filed before Form N-CSR replaced it
N-30B-2	Other periodic or interim report to shareholders
N-Q	Schedule of portfolio holdings for the first and third fiscal quarters
N-PX	Annual report of proxy voting record
NSAR-A	Semiannual report of a management investment company
NSAR-B	Annual report of a management investment company
NSAR-U	Annual report of a unit investment trust
N-23C3A	Notification of a periodic repurchase offer by an interval fund
<i>Institutional investment managers, large traders, and transfer agents</i>	
13F-HR	Quarterly report of equity holdings by an institutional investment manager
13F-NT	Quarterly notice by a manager whose holdings are reported by another manager
13H-A	Annual update of a large trader's registration
TA-2	Annual report of a transfer agent's activities
<i>Foreign governments and development banks</i>	
18-K	Annual report of a foreign government or political subdivision
ANRLRPT	Annual report of a development bank
QRTLYRPT	Quarterly report of a development bank

Table A.4: Scheduled form types

Note: Scheduled form types are those whose filing *dates* are known in advance. The exact acceptance *time* remains unknown until the filing is submitted. Amendments of these forms are scheduled as well. Unscheduled form types are the remaining EXCL forms. Form types are grouped by the type of filer, and the descriptions follow the SEC's EDGAR form descriptions.

<i>Panel A: by calendar year</i>			
Year	N	Mean Q_{30}^{pre} (bps)	Mean Q_{30}^{post} (bps)
2003	256	-7.97	-0.31
2004	1,719	-1.42	-6.42
2005	2,796	-5.13	-0.24
2006	5,420	1.22	0.08
2007	6,377	0.65	-0.81
2008	5,929	-0.09	-0.84
2009	4,765	-0.35	2.18
2010	4,136	-2.52	0.94
2011	3,775	-2.29	1.78
2012	3,028	-1.64	-1.18
2013	2,651	1.41	-0.45
2014	2,857	1.13	-0.54
2015	2,655	-0.74	1.43
2016	2,287	0.93	1.58
2017	2,088	1.68	1.57
2018	2,183	-0.38	0.88
2019	2,267	1.60	-0.12
2020	2,753	-3.21	-5.75
2021	2,368	0.37	-1.53
2022	2,319	-1.06	-0.76
2023	2,803	-0.26	-1.30
2024	3,137	-0.59	-0.27
<i>Panel B: by TAQ product era</i>			
Era	N	5–1	
2013–2024	30,368	5.89 bps ($t=3.37$)	
2003–2012	38,201	10.20 bps ($t=5.43$)	

Table A.5: Filings per year and by *TAQ* product era, EXCL estimation sample

Note: Panel A reports filings and mean returns by calendar year. Panel B splits the sample at 1 January 2013 and reports the 5–1 in each era; the underlying *TAQ* product changes across that break.

III. Single Sort Results

Q_{30}^{pre}	bin	Mean Q_{30}^{post} (bps)	t	Std. err. (bps)
	5	3.75***	6.65	0.56
	4	0.85***	2.70	0.32
	3	0.47*	1.75	0.27
	2	-0.14	-0.45	0.30
	1	-0.44	-0.77	0.57
5–1		4.18***	5.22	0.80
Total Observations N = 145,513				

Table A.6: Single sort, all-form-type sample, 30-minute windows

Note: The sample includes all form types accepted by EDGAR between September 2003 and December 2024. Filings are restricted to informative firm-quarters (Section 3.3). Pre- and post-acceptance returns are measured over 30-minute windows around the EDGAR acceptance timestamp of each filing. Quintile means report one-sample t -statistics, and the 5–1 reports the Welch two-sample statistic. To indicate significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Q_{30}^{pre}	bin	Mean Q_{30}^{post} (bps)	t	Std. err. (bps)
	5	1.97	1.21	1.63
	4	1.82*	1.65	1.10
	3	-0.36	-0.44	0.82
	2	-1.74*	-1.66	1.05
	1	-3.71**	-2.23	1.66
5–1		5.68**	2.44	2.33
Total Observations N = 17,072				

Table A.7: Single sort, SCHE, 30-minute windows

Note: The sample includes scheduled EXCL form types (SCHE) accepted by EDGAR between September 2003 and December 2024. Filings are restricted to informative firm-quarters (Section 3.3). Pre- and post-acceptance returns are measured over 30-minute windows around the EDGAR acceptance timestamp of each filing. Quintile means report one-sample t -statistics, and the 5–1 reports the Welch two-sample statistic. To indicate significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Q_{30}^{pre}	bin	Mean Q_{30}^{post} (bps)	t	Std. err. (bps)
	5	5.29***	4.86	1.09
	4	-0.54	-0.95	0.56
	3	-0.57	-1.22	0.47
	2	-0.95*	-1.80	0.53
	1	-4.51***	-4.05	1.11
5–1		9.80***	6.30	1.56
Total Observations N = 51,497				

Table A.8: Single sort, UNSC, 30-minute windows

Note: The sample includes unscheduled EXCL form types (UNSC) accepted by EDGAR between September 2003 and December 2024. Filings are restricted to informative firm-quarters (Section 3.3). Pre- and post-acceptance returns are measured over 30-minute windows around the EDGAR acceptance timestamp of each filing. Quintile means report one-sample t -statistics, and the 5–1 reports the Welch two-sample statistic. To indicate significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Q_{30}^{pre}	bin	Mean Q_{30}^{post} (bps)	t	Std. err. (bps)
	5	9.64***	4.97	1.94
	4	0.71	0.68	1.05
	3	-0.71	-0.86	0.82
	2	-0.39	-0.41	0.95
	1	-5.88***	-2.86	2.05
5–1		15.52***	5.49	2.83
Total Observations N = 23,658				

Table A.9: Single sort, Form 8-K, 30-minute windows

Note: The sample includes EXCL Form 8-K filings and their amendments and variants (such as Form 8-K12G3), accepted by EDGAR between September 2003 and December 2024. Filings are restricted to informative firm-quarters (Section 3.3). Pre- and post-acceptance returns are measured over 30-minute windows around the EDGAR acceptance timestamp of each filing. Quintile means report one-sample t -statistics, and the 5–1 reports the Welch two-sample statistic. To indicate significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Q_{30}^{pre}	bin	Mean Q_{30}^{post} (bps)	t	Std. err. (bps)
	5	1.67	0.71	2.35
	4	2.74	1.56	1.75
	3	0.84	0.69	1.21
	2	-2.42	-1.51	1.60
	1	-5.86**	-2.36	2.48
	5–1	7.52**	2.20	3.41
Total Observations N = 9,930				

Table A.10: Single sort, Form 10-Q, 30-minute windows

Note: The sample includes EXCL Form 10-Q filings and their amendments, accepted by EDGAR between September 2003 and December 2024. Filings are restricted to informative firm-quarters (Section 3.3). Pre- and post-acceptance returns are measured over 30-minute windows around the EDGAR acceptance timestamp of each filing. Quintile means report one-sample t -statistics, and the 5–1 reports the Welch two-sample statistic. To indicate significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Q_{30}^{pre}	bin	Mean Q_{30}^{post} (bps)	t	Std. err. (bps)
	5	2.32	0.58	3.99
	4	-0.30	-0.14	2.11
	3	-0.02	-0.01	1.77
	2	-0.44	-0.22	2.03
	1	-0.89	-0.25	3.53
	5–1	3.21	0.60	5.32
Total Observations N = 2,841				

Table A.11: Single sort, Form 10-K, 30-minute windows

Note: The sample includes EXCL Form 10-K filings and their amendments, accepted by EDGAR between September 2003 and December 2024. Filings are restricted to informative firm-quarters (Section 3.3). Pre- and post-acceptance returns are measured over 30-minute windows around the EDGAR acceptance timestamp of each filing. Quintile means report one-sample t -statistics, and the 5–1 reports the Welch two-sample statistic. To indicate significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Q_{30}^{pre}	bin	Mean Q_{30}^{post} (bps)	t	Std. err. (bps)
	5	0.81	0.80	1.01
	4	-0.30	-0.56	0.54
	3	-0.75	-1.59	0.48
	2	-1.60***	-2.93	0.55
	1	-2.90***	-2.97	0.98
	5–1	3.71***	2.63	1.41
Total Observations N = 32,140				

Table A.12: Single sort, EXCL excluding Forms 8-K, 10-K, and 10-Q, 30-minute windows

Note: The sample includes the EXCL form types other than Forms 8-K, 10-K, and 10-Q, each with their amendments, accepted by EDGAR between September 2003 and December 2024. Filings are restricted to informative firm-quarters (Section 3.3). Pre- and post-acceptance returns are measured over 30-minute windows around the EDGAR acceptance timestamp of each filing. Quintile means report one-sample t -statistics, and the 5–1 reports the Welch two-sample statistic. To indicate significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

IV. Panel Regressions with Filing-Date Fixed Effects

	Post-acceptance return				
	No control (1)	Size (2)	Returns (3)	Views (4)	All (5)
Pre-acceptance return	0.0524*** (3.82)	0.0524*** (3.81)	0.0521*** (3.78)	0.0568*** (3.41)	0.0563*** (3.36)
Log firm size		0.0079 (0.25)			0.0236 (0.44)
Return - overnight			-0.0118 (-1.21)		-0.0146 (-1.28)
Return - 1 day			-0.0219** (-2.48)		-0.0250** (-2.47)
Return - 7 days			-0.0060 (-0.56)		-0.0060 (-0.48)
Return - 30 days			0.0038 (0.34)		0.0029 (0.22)
Return - 180 days			-0.0059 (-0.83)		-0.0070 (-0.90)
Return - 365 days			-0.0061 (-0.61)		-0.0187 (-1.52)
Total views, 30 days before				0.0019 (0.87)	0.0022 (1.04)
Institutional views dummy, 30 days before				0.0039 (0.71)	0.0035 (0.65)
Institutional views, 30 days before				-0.0025 (-0.62)	-0.0027 (-0.65)
Observations	67,977	67,977	67,976	49,423	49,422
R ²	0.1483	0.1483	0.1490	0.1519	0.1529
Firm fixed effects	✓	✓	✓	✓	✓
Date fixed effects	✓	✓	✓	✓	✓
Two-way clustered <i>t</i> -statistics	✓	✓	✓	✓	✓

Table A.13: Fixed-effect regression of the post-acceptance return on the pre-acceptance return, EXCL form types, firm and filing-date fixed effects

Note: The table repeats Table 2 with firm and filing-date fixed effects in place of firm and month fixed effects and with standard errors clustered two-way by firm and filing date. Filings that are the only observation of their firm or of their filing date drop from the estimation. The sample, controls, and standardization match Table 2. *t*-statistics are displayed below coefficient estimates. To indicate significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

	Post-acceptance return				
	No control (1)	Size (2)	Returns (3)	Views (4)	All (5)
Pre-acceptance return	0.0517*** (3.62)	0.0517*** (3.61)	0.0514*** (3.58)	0.0560*** (3.30)	0.0559*** (3.29)
Log firm size		0.0004 (0.06)			-0.0063 (-0.73)
Return - overnight			-0.0142 (-1.37)		-0.0171 (-1.38)
Return - 1 day			-0.0312*** (-3.25)		-0.0344*** (-3.16)
Return - 7 days			-0.0024 (-0.19)		0.0018 (0.12)
Return - 30 days			0.0047 (0.42)		0.0052 (0.41)
Return - 180 days			-0.0120 (-1.61)		-0.0105 (-1.44)
Return - 365 days			-0.0021 (-0.27)		-0.0055 (-0.79)
Total views, 30 days before				-0.0007 (-0.32)	0.0001 (0.04)
Institutional views dummy, 30 days before				0.0013 (0.26)	0.0020 (0.37)
Institutional views, 30 days before				-0.0014 (-0.43)	-0.0010 (-0.29)
Observations	68,497	68,497	68,496	49,925	49,924
R ²	0.0814	0.0814	0.0826	0.0788	0.0801
Date fixed effects	✓	✓	✓	✓	✓
Date-clustered <i>t</i> -statistics	✓	✓	✓	✓	✓

Table A.14: Fixed-effect regression of the post-acceptance return on the pre-acceptance return, EXCL form types, filing-date fixed effects

Note: The table repeats Table 2 with filing-date fixed effects in place of firm and month fixed effects and with standard errors clustered by filing date. Filing dates with a single filing drop from the estimation. The sample, controls, and standardization match Table 2. *t*-statistics are displayed below coefficient estimates. To indicate significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

V. Panel Regression with Fixed Effects and S&P 500 Adjustment

	No control (1)	Post-acceptance return (adj.)			
		Size (2)	Returns (3)	Views (4)	All (5)
Pre-acceptance return (adj.)	0.0670*** (4.84)	0.0670*** (4.83)	0.0668*** (4.82)	0.0732*** (4.32)	0.0729*** (4.28)
Log firm size		0.0123 (0.42)			0.0257 (0.52)
Return - overnight			-0.0004 (-0.04)		-0.0037 (-0.30)
Return - 1 day			-0.0203*** (-2.68)		-0.0219** (-2.42)
Return - 7 days			-0.0042 (-0.43)		-0.0018 (-0.15)
Return - 30 days			0.0031 (0.30)		-0.0015 (-0.11)
Return - 180 days			-0.0044 (-0.61)		-0.0041 (-0.56)
Return - 365 days			-0.0118 (-1.05)		-0.0243* (-1.96)
Total views, 30 days before				-0.0001 (-0.07)	0.0005 (0.31)
Institutional views dummy, 30 days before				0.0055 (0.99)	0.0053 (0.95)
Institutional views, 30 days before				-0.0025 (-0.64)	-0.0029 (-0.71)
Observations	68,023	68,023	68,022	49,430	49,429
R ²	0.0800	0.0800	0.0807	0.0864	0.0872
Firm fixed effects	✓	✓	✓	✓	✓
Month fixed effects	✓	✓	✓	✓	✓
Two-way clustered <i>t</i> -statistics	✓	✓	✓	✓	✓

Table A.15: Fixed-effect regression of the post-acceptance return on the pre-acceptance return, EXCL form types, market-adjusted

Note: The table repeats Table 2 after subtracting the concurrent S&P 500 return from both the pre- and post-acceptance returns. The sample, controls, firm and month fixed effects, standardization, and two-way clustering by firm and calendar month match Table 2. Filings that are the only observation of their firm or of their calendar month drop from the estimation. *t*-statistics are displayed below coefficient estimates. To indicate significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

VI. Panel Regression after Removing Preceding Firm-Specific News

Panel A: No firm-specific news in the six calendar days before the filing date

	No control	Post-acceptance return			All
	(1)	Size	Returns	Views	(5)
Pre-acceptance return	0.0753*** (4.20)	0.0753*** (4.20)	0.0746*** (4.16)	0.0803*** (4.09)	0.0796*** (4.03)
Log firm size		-0.0064 (-0.19)			0.0101 (0.25)
Return - overnight			-0.0031 (-0.26)		-0.0035 (-0.28)
Return - 1 day			-0.0270*** (-2.95)		-0.0250*** (-2.65)
Return - 7 days			-0.0094 (-0.85)		-0.0124 (-1.07)
Return - 30 days			-0.0015 (-0.13)		-0.0027 (-0.21)
Return - 180 days			-0.0026 (-0.34)		-0.0013 (-0.16)
Return - 365 days			-0.0149 (-1.17)		-0.0169 (-1.27)
Total views, 30 days before				-0.0010 (-0.33)	-0.0010 (-0.29)
Institutional views dummy, 30 days before				0.0059 (0.93)	0.0060 (0.94)
Institutional views, 30 days before				-0.0023 (-0.35)	-0.0025 (-0.37)
Observations	38,344	38,344	38,343	35,427	35,426
R ²	0.0987	0.0987	0.1000	0.0989	0.1003
Firm fixed effects	✓	✓	✓	✓	✓
Month fixed effects	✓	✓	✓	✓	✓
Two-way clustered <i>t</i> -statistics	✓	✓	✓	✓	✓

Table A.16: Fixed-effect regression of the post-acceptance return on the pre-acceptance return, EXCL form types, no preceding *RavenPack* news

Note: The table repeats Table 2 after removing filings with a firm-specific *RavenPack* news event (Section 4.4.1) in the six calendar days before the filing date. The controls, firm and month fixed effects, standardization, and two-way clustering by firm and calendar month match Table 2. Filings that are the only observation of their firm or of their calendar month drop from the estimation. *t*-statistics are displayed below coefficient estimates. To indicate significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Panel B: No firm-specific news in the six calendar days before the filing date or earlier on the filing date

	Post-acceptance return				
	No control (1)	Size (2)	Returns (3)	Views (4)	All (5)
Pre-acceptance return	0.0588*** (3.70)	0.0587*** (3.70)	0.0580*** (3.66)	0.0622*** (3.59)	0.0613*** (3.53)
Log firm size		0.0069 (0.20)			0.0340 (0.83)
Return - overnight			-0.0018 (-0.15)		-0.0029 (-0.23)
Return - 1 day			-0.0253*** (-3.20)		-0.0228*** (-2.86)
Return - 7 days			-0.0118 (-1.08)		-0.0141 (-1.19)
Return - 30 days			0.0066 (0.61)		0.0029 (0.25)
Return - 180 days			-0.0039 (-0.51)		-0.0021 (-0.27)
Return - 365 days			-0.0193 (-1.48)		-0.0234* (-1.68)
Total views, 30 days before				0.0004 (0.07)	0.0006 (0.10)
Institutional views dummy, 30 days before				0.0061 (0.89)	0.0060 (0.88)
Institutional views, 30 days before				-0.0045 (-0.65)	-0.0047 (-0.67)
Observations	35,372	35,372	35,371	32,730	32,729
R ²	0.1021	0.1021	0.1034	0.1013	0.1026
Firm fixed effects	✓	✓	✓	✓	✓
Month fixed effects	✓	✓	✓	✓	✓
Two-way clustered <i>t</i> -statistics	✓	✓	✓	✓	✓

Table A.16: Fixed-effect regression of the post-acceptance return on the pre-acceptance return, EXCL form types, no preceding *RavenPack* news (continued)

Note: Panel B removes, in addition to the filings removed in Panel A, all filings with a firm-specific *RavenPack* news event between midnight and the EDGAR acceptance timestamp on the filing date. The specifications, the standardization, and the two-way clustering by firm and calendar month match Panel A. Filings that are the only observation of their firm or of their calendar month drop from the estimation. *t*-statistics are displayed below coefficient estimates. To indicate significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

VII. Filings Versus Placebo

	Estimate	Std. err.	<i>t</i>
<i>Panel A. Single-sort difference</i>			
Filing 5–1 (bps)	8.86***	1.31	6.78
Placebo 5–1 (bps)	0.72	1.21	0.59
Filing-minus-placebo 5–1 (bps)	8.14***	1.78	4.57
<i>Panel B. Continuation slope (pooled regression)</i>			
Filing continuation slope	0.0652***	0.0133	4.90
Placebo continuation slope	-0.0189	0.0170	-1.11
Filing-minus-placebo slope	0.0841***	0.0216	3.90
Total Observations N = 68,569 (filings); 98,374 (placebo)			

Table A.17: Filings versus placebo: the identifying difference

Note: The sample (EXCL form types) includes all form types excluding 3/4/5/13D/13G accepted by EDGAR between September 2003 and December 2024 and the placebo events described in Section 5.1. Filings and placebo events are restricted to informative firm-quarters (Section 3.3). Pre- and post-acceptance returns are measured over 30-minute windows around the EDGAR acceptance timestamp of each filing and around the event time of each placebo event. Panel A compares the single-sort 5–1, with quintiles formed within each sample arm (genuine filings versus random-draw placebos); its *t*-statistics are Welch two-sample statistics, and the difference tests the difference of the two 5–1s. Panel B compares continuation slopes from a pooled regression of Q_{30}^{post} on Q_{30}^{pre} interacted with a filing indicator, without fixed effects or controls. The filing slope is the sum of the placebo slope and the difference, with its standard error from the same regression. Panel B’s standard errors are clustered two-way by firm and event date. To indicate significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

VIII. Alternative Event Price

Event-price definition	N	Filing		Placebo		Difference	
		5–1	<i>t</i>	5–1	<i>t</i>	5–1	<i>t</i>
Baseline: post from $[0, 3)$	68,569	8.86***	6.78	0.72	0.59	8.14***	4.57
Alternative: post from $[-3, 0)$	72,381	11.84***	8.69	3.98***	3.26	7.86***	4.30
Baseline, market-adjusted	68,537	7.39***	5.93	-0.08	-0.07	7.46***	4.36
Alternative, market-adjusted	72,347	9.95***	7.65	2.09*	1.77	7.85***	4.47

Table A.18: Filings versus placebo under alternative event-price definitions

Note: The sample (EXCL form types) includes all form types excluding 3/4/5/13D/13G accepted by EDGAR between September 2003 and December 2024 and the placebo events described in Section 5.1. Filings and placebo events are restricted to informative firm-quarters (Section 3.3). Pre- and post-acceptance returns are measured over 30-minute windows around the EDGAR acceptance timestamp of each filing and around the event time of each placebo event. Each row re-forms the pre-acceptance-return quintiles within each arm under a different definition of the event price and reports the 5–1 of the post-acceptance return, in basis points, for EXCL filings, for the placebo, and for their difference. The baseline follows the definition in Section 3.2: the pre-acceptance return ends at $[-3, 0)$ and the post-acceptance return starts at $[0, 3)$, so the two returns share no price. The alternative starts the post-acceptance return at $[-3, 0)$, where the pre-acceptance return ends, so the two returns share that price. Market-adjusted rows subtract the concurrent S&P 500 return from both legs. The *t*-statistics are Welch two-sample statistics; the difference tests the difference of the two 5–1s. To indicate significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

IX. Pre-Event Order Flow

Q_{30}^{pre}	bin	Mean X_{LR}	t	Std. err.
	5	0.1253***	54.93	0.0023
	4	0.0584***	26.41	0.0022
	3	0.0015	0.62	0.0024
	2	-0.0557***	-24.62	0.0023
	1	-0.1205***	-51.96	0.0023
	5–1	0.2458***	75.56	0.0033
Total Observations N = 68,563				

Table A.19: Lee–Ready order imbalance by Q_{30}^{pre} quintile, EXCL form types

Note: The sample (EXCL form types) includes all form types excluding 3/4/5/13D/13G accepted by EDGAR between September 2003 and December 2024. Filings are restricted to informative firm-quarters (Section 3.3). Pre-acceptance returns and order imbalances are measured over the 30 minutes before the EDGAR acceptance timestamp of each filing. The outcome is the filing-level order imbalance X_{LR} of equation (5). The sample requires both return legs to be non-missing, as elsewhere in the paper; Section 6.1 reports the unrestricted sample. To indicate significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Internet Appendix

IA.A. Sample Selection

#	Filter	Filings	Dropped	% of stage 1
1	EDGAR filings accepted September 2003 to December 2024, CIK–CUSIP link valid at the event date, WRDS company-name match, <i>CRSP</i> daily record, accepted 9:30 a.m.–4:00 p.m. ET	2,064,810	—	100.0
2	Accepted 10:00 a.m.–3:30 p.m. ET	1,725,228	339,582	83.6
3	<i>CRSP</i> SHRCD 10 or 11 and EXCD 1–4 at the event date	1,464,135	261,093	70.9
4	Drop same-CUSIP same-day filings accepted within 30 minutes of each other	698,481	765,654	33.8
5	Matched to a <i>TAQ</i> symbol at the event date, at least one trade in the ± 30 -minute window	616,681	81,800	29.9
6	Exclude Forms 3, 4, 5, SC 13D, SC 13G, and amendments (EXCL form types)	287,704	328,977	13.9
7	Valid pre-acceptance return Q_{30}^{pre}	177,707	109,997	8.6
8	Valid post-acceptance return Q_{30}^{post}	157,154	20,553	7.6
9	In a firm-quarter with an operating-cash-flow forecast error	149,439	7,715	7.2
10	Firm-quarter forecast error above the median (estimation sample)	68,569	80,870	3.3

Table IA.1: Sample selection

Note: Stages 1 to 5 are constructed from the WRDS event sheets and *TAQ* extraction (Section 3.1), and stages 6 to 10 from the resulting filing panel. Because a CIK can link to more than one CUSIP, each count is the number of unique filings. Stage 1 uses the WRDS SEC Analytics Suite, including its CIK–CUSIP linking table and company names, and requires a *CRSP* daily record for the linked CUSIP on the event date. Stage 2 ensures full 30-minute windows within trading hours. Stage 3 keeps common shares, SHRCD 10 and 11, traded on NYSE, AMEX, NASDAQ, or Arca, *CRSP* exchange codes 1–4, at the event date. Stage 4 drops every filing that has another filing for the same CUSIP on the same day within 30 minutes of its acceptance timestamp, including exactly 30 minutes. Stages 9 and 10 apply the cash-flow filter of Section 3.3: stage 9 keeps filings in a firm-quarter with an operating-cash-flow forecast error, and stage 10 keeps filings in informative firm-quarters, whose absolute forecast error exceeds the median. The estimation sample at stage 10 is the sample used in every 30-minute EXCL table in the paper, and stage 8 is the all-firm-quarters sample of Table A.1.

IA.B. Fifteen-Second Price Path, All Form Types

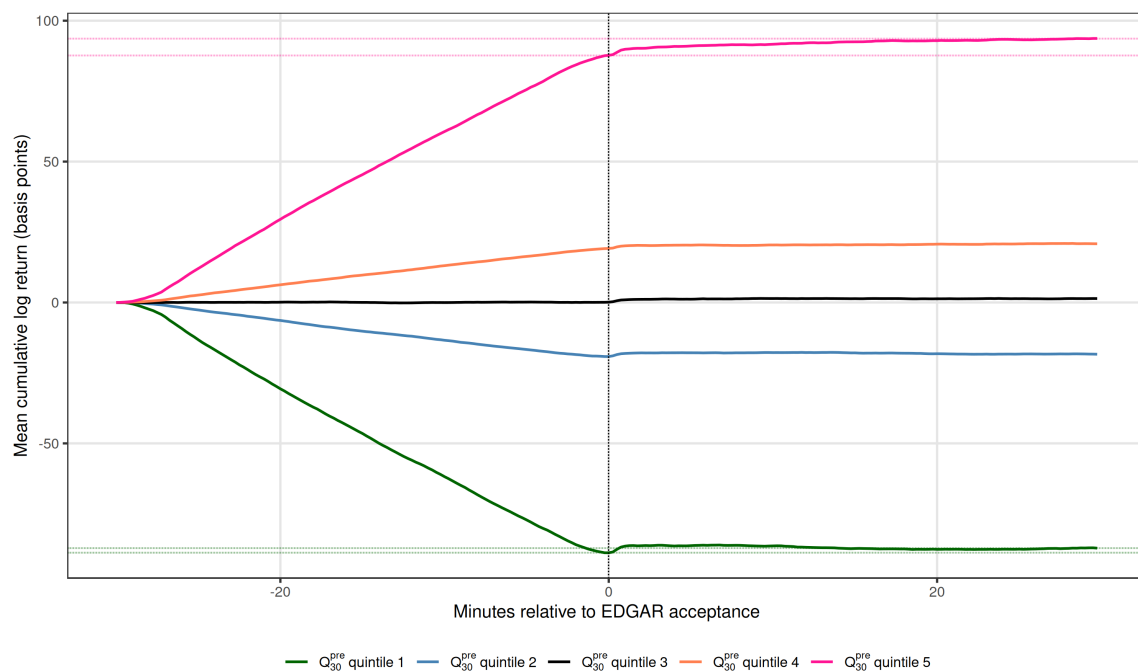


Figure IA.1: Cumulative returns around EDGAR acceptance by pre-acceptance-return quintile, all-form-type sample

Note: The figure repeats Figure 2 for all form types accepted by EDGAR between September 2003 and December 2024 in informative firm-quarters (Section 3.3). The quintiles, the plotted returns, and the reference lines are defined as in Figure 2.

IA.C. Clustered Inference for the Baseline Single Sort

Q_{30}^{pre}	<i>bin</i>	Mean Q_{30}^{post} (bps)	<i>t</i> -statistic		
			i.i.d.	Cl. month	Cl. firm+month
5		4.42***	4.84	5.04	4.91
4		0.07	0.15	0.14	0.14
3		-0.46	-1.13	-1.08	-1.11
2		-1.07**	-2.26	-2.19	-2.11
1		-4.44***	-4.75	-4.92	-5.30
5–1		8.86***	6.78	6.60	6.71
Total Observations N = 68,569					
Event dates = 5,164; event months = 256; firms = 3,595					
Month-block bootstrap 95% CI for 5–1: [6.19, 11.57] bps					
Within-month randomization-inference <i>p</i> for 5–1: 0.0010					

Table IA.2: Single sort with clustered inference, EXCL, 30-minute windows

Note: The sample (EXCL form types) includes all form types excluding 3/4/5/13D/13G accepted by EDGAR between September 2003 and December 2024. Filings are restricted to informative firm-quarters (Section 3.3). Pre- and post-acceptance returns are measured over 30-minute windows around the EDGAR acceptance timestamp of each filing. Quintile portfolios are formed on Q_{30}^{pre} with pooled full-sample breakpoints, and the outcome is Q_{30}^{post} in basis points. Point estimates match Table 1; only the standard errors differ. “Cl. month” clusters on the calendar month of the EDGAR acceptance date, and “Cl. firm+month” clusters two-way on firm and that month. The month-block bootstrap resamples those calendar months with replacement, and the randomization test permutes quintile labels within each month. Stars follow the two-way clustered *p*-value. To indicate significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

IA.D. Portfolio Differences by Form-Type Group

Q_{30}^{pre} bin j	Q_{30}^{pre} bin i	Mean (bps)				t				Std. err. (bps)			
		5	4	3	2	5	4	3	2	5	4	3	2
	4	4.35***	—	—	—	4.17	—	—	—	1.04	—	—	—
	3	4.88***	0.53	—	—	4.89	0.82	—	—	1.00	0.65	—	—
	2	5.49***	1.14*	0.61	—	5.34	1.65	0.99	—	1.03	0.69	0.62	—
	1	8.86***	4.51***	3.98***	3.37***	6.78	4.25	3.91	3.21	1.31	1.06	1.02	1.05
Total Observations N = 68,569													

Table IA.3: Portfolio return difference, EXCL, 30-minute windows

Note: The sample (EXCL form types) includes all form types excluding 3/4/5/13D/13G accepted by EDGAR between September 2003 and December 2024. Filings are restricted to informative firm-quarters (Section 3.3). Pre- and post-acceptance returns are measured over 30-minute windows around the EDGAR acceptance timestamp of each filing. Differences report the Welch two-sample statistic. To indicate significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Q_{30}^{pre} bin j	Q_{30}^{pre} bin i	Mean (bps)				t				Std. err. (bps)			
		5	4	3	2	5	4	3	2	5	4	3	2
	4	2.89***	—	—	—	4.48	—	—	—	0.65	—	—	—
	3	3.28***	0.39	—	—	5.26	0.94	—	—	0.62	0.41	—	—
	2	3.88***	0.99**	0.60	—	6.09	2.27	1.50	—	0.64	0.44	0.40	—
	1	4.18***	1.29**	0.90	0.30	5.22	1.98	1.44	0.47	0.80	0.65	0.63	0.64
Total Observations N = 145,513													

Table IA.4: Portfolio return difference, all-form-type sample, 30-minute windows

Note: The sample includes all form types accepted by EDGAR between September 2003 and December 2024. Filings are restricted to informative firm-quarters (Section 3.3). Pre- and post-acceptance returns are measured over 30-minute windows around the EDGAR acceptance timestamp of each filing. Differences report the Welch two-sample statistic. To indicate significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Q_{30}^{pre} bin j	Q_{30}^{pre} bin i	Mean (bps)				t				Std. err. (bps)			
		5	4	3	2	5	4	3	2	5	4	3	2
	4	0.15	—	—	—	0.08	—	—	—	1.97	—	—	—
	3	2.32	2.17	—	—	1.27	1.58	—	—	1.82	1.37	—	—
	2	3.71*	3.56**	1.38	—	1.91	2.34	1.04	—	1.94	1.52	1.33	—
	1	5.68**	5.53***	3.35*	1.97	2.44	2.77	1.81	1.00	2.33	1.99	1.85	1.96
Total Observations N = 17,072													

Table IA.5: Portfolio return difference, SCHE, 30-minute windows

Note: The sample includes scheduled EXCL form types (SCHE) accepted by EDGAR between September 2003 and December 2024. Filings are restricted to informative firm-quarters (Section 3.3). Pre- and post-acceptance returns are measured over 30-minute windows around the EDGAR acceptance timestamp of each filing. Differences report the Welch two-sample statistic. To indicate significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

		Mean (bps)				t				Std. err. (bps)			
		5	4	3	2	5	4	3	2	5	4	3	2
Q_{30}^{pre}	$bin\ j$	$Q_{30}^{pre}\ bin\ i$											
	4	5.82***	—	—	—	4.76	—	—	—	1.22	—	—	—
	3	5.86***	0.03	—	—	4.95	0.04	—	—	1.18	0.73	—	—
	2	6.24***	0.42	0.39	—	5.16	0.54	0.55	—	1.21	0.77	0.71	—
	1	9.80***	3.98***	3.95***	3.56***	6.30	3.18	3.27	2.88	1.56	1.25	1.21	1.23
Total Observations N = 51,497													

Table IA.6: Portfolio return difference, UNSC, 30-minute windows

Note: The sample includes unscheduled EXCL form types (UNSC) accepted by EDGAR between September 2003 and December 2024. Filings are restricted to informative firm-quarters (Section 3.3). Pre- and post-acceptance returns are measured over 30-minute windows around the EDGAR acceptance timestamp of each filing. Differences report the Welch two-sample statistic. To indicate significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

		Mean (bps)				t				Std. err. (bps)			
		5	4	3	2	5	4	3	2	5	4	3	2
Q_{30}^{pre}	$bin\ j$	$Q_{30}^{pre}\ bin\ i$											
	4	8.93***	—	—	—	4.05	—	—	—	2.21	—	—	—
	3	10.35***	1.42	—	—	4.91	1.06	—	—	2.11	1.33	—	—
	2	10.04***	1.10	-0.31	—	4.64	0.78	-0.25	—	2.16	1.42	1.26	—
	1	15.52***	6.59***	5.17**	5.48**	5.49	2.86	2.34	2.42	2.83	2.30	2.21	2.26
Total Observations N = 23,658													

Table IA.7: Portfolio return difference, Form 8-K, 30-minute windows

Note: The sample includes EXCL Form 8-K filings and their amendments and variants (such as Form 8-K12G3), accepted by EDGAR between September 2003 and December 2024. Filings are restricted to informative firm-quarters (Section 3.3). Pre- and post-acceptance returns are measured over 30-minute windows around the EDGAR acceptance timestamp of each filing. Differences report the Welch two-sample statistic. To indicate significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

		Mean (bps)				t				Std. err. (bps)			
		5	4	3	2	5	4	3	2	5	4	3	2
Q_{30}^{pre}	$bin\ j$	$Q_{30}^{pre}\ bin\ i$											
	4	-1.07	—	—	—	-0.37	—	—	—	2.93	—	—	—
	3	0.83	1.90	—	—	0.31	0.89	—	—	2.64	2.13	—	—
	2	4.09	5.17**	3.26	—	1.44	2.17	1.62	—	2.84	2.38	2.01	—
	1	7.52**	8.60***	6.70**	3.43	2.20	2.83	2.43	1.16	3.41	3.04	2.76	2.95
Total Observations N = 9,930													

Table IA.8: Portfolio return difference, Form 10-Q, 30-minute windows

Note: The sample includes EXCL Form 10-Q filings and their amendments, accepted by EDGAR between September 2003 and December 2024. Filings are restricted to informative firm-quarters (Section 3.3). Pre- and post-acceptance returns are measured over 30-minute windows around the EDGAR acceptance timestamp of each filing. Differences report the Welch two-sample statistic. To indicate significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

		Mean (bps)				t				Std. err. (bps)			
		5	4	3	2	5	4	3	2	5	4	3	2
Q_{30}^{pre}	$bin\ j$	Q_{30}^{pre}	$bin\ i$										
	4	2.61	—	—	—	0.58	—	—	—	4.51	—	—	—
	3	2.34	-0.27	—	—	0.54	-0.10	—	—	4.36	2.76	—	—
	2	2.76	0.14	0.41	—	0.62	0.05	0.15	—	4.47	2.93	2.69	—
	1	3.21	0.60	0.87	0.45	0.60	0.15	0.22	0.11	5.32	4.11	3.95	4.07
Total Observations N = 2,841													

Table IA.9: Portfolio return difference, Form 10-K, 30-minute windows

Note: The sample includes EXCL Form 10-K filings and their amendments, accepted by EDGAR between September 2003 and December 2024. Filings are restricted to informative firm-quarters (Section 3.3). Pre- and post-acceptance returns are measured over 30-minute windows around the EDGAR acceptance timestamp of each filing. Differences report the Welch two-sample statistic. To indicate significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

		Mean (bps)				t				Std. err. (bps)			
		5	4	3	2	5	4	3	2	5	4	3	2
Q_{30}^{pre}	$bin\ j$	Q_{30}^{pre}	$bin\ i$										
	4	1.11	—	—	—	0.96	—	—	—	1.15	—	—	—
	3	1.56	0.45	—	—	1.39	0.63	—	—	1.12	0.72	—	—
	2	2.41**	1.30*	0.84	—	2.09	1.69	1.16	—	1.15	0.77	0.72	—
	1	3.71***	2.60**	2.14**	1.30	2.63	2.32	1.97	1.16	1.41	1.12	1.09	1.12
Total Observations N = 32,140													

Table IA.10: Portfolio return difference, EXCL excluding Forms 8-K, 10-K, and 10-Q, 30-minute windows

Note: The sample includes the EXCL form types other than Forms 8-K, 10-K, and 10-Q, each with their amendments, accepted by EDGAR between September 2003 and December 2024. Filings are restricted to informative firm-quarters (Section 3.3). Pre- and post-acceptance returns are measured over 30-minute windows around the EDGAR acceptance timestamp of each filing. Differences report the Welch two-sample statistic. To indicate significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

IA.E. Panel Regression with Fixed Effects, Placebo Sample

	No control (1)	Post-acceptance return			All (5)
		Size (2)	Returns (3)	Views (4)	
Pre-acceptance return	-0.0175 (-0.90)	-0.0176 (-0.90)	-0.0162 (-0.86)	0.0246 (1.43)	0.0243 (1.39)
Log firm size		0.0088 (0.49)			0.0261 (1.10)
Return - overnight			-0.0444 (-1.55)		0.0014 (0.06)
Return - 1 day			-0.0048 (-0.33)		0.0223 (1.10)
Return - 7 days			0.0008 (0.07)		-0.0595 (-1.31)
Return - 30 days			0.0006 (0.07)		0.0311* (1.80)
Return - 180 days			0.0044 (0.66)		-0.0062 (-0.62)
Return - 365 days			-0.0106 (-1.49)		-0.0030 (-0.38)
Total views, 30 days before				0.0012 (1.07)	0.0012 (1.14)
Institutional views dummy, 30 days before				-0.0040 (-0.99)	-0.0041 (-1.01)
Institutional views, 30 days before				0.0009 (0.36)	0.0011 (0.44)
Observations	97,934	97,934	97,934	51,619	51,619
R ²	0.1035	0.1035	0.1058	0.0862	0.0875
Firm fixed effects	✓	✓	✓	✓	✓
Month fixed effects	✓	✓	✓	✓	✓
Two-way clustered <i>t</i> -statistics	✓	✓	✓	✓	✓

Table IA.11: Fixed-effect regression of the post-acceptance return on the pre-acceptance return, placebo sample

Note: The table repeats Table 2 on the placebo events described in Section 5.1, drawn between September 2003 and December 2024 and restricted to informative firm-quarters (Section 3.3). Pre- and post-acceptance returns are measured over 30-minute windows around each placebo event time. The controls, firm and month fixed effects, standardization, and two-way clustering by firm and calendar month match Table 2. Placebo events that are the only observation of their firm or of their calendar month drop from the estimation. *t*-statistics are displayed below coefficient estimates. To indicate significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

IA.F. Placebo Draws that Coincide with a Filing

<i>Panel A: coincidence rate</i>				
Placebo draws on a day the firm actually filed	3,276 (3.33%)			
of which within 60 minutes of that filing	1,160 (1.18%)			
of which within 30 minutes of that filing	606 (0.62%)			
Placebo draws on that day or an adjacent day	7,724 (7.85%)			
<i>Panel B: placebo 5–1 after removing coincident draws</i>				
Subsample	N	5–1 (bps)	t	p
All placebo draws	98,374	0.72	0.59	0.5543
Drop draws on a filing day	95,098	0.46	0.37	0.7102
Drop draws on or next to a filing day	90,650	0.64	0.50	0.6157
Drop draws within 60 minutes of a filing	97,214	0.73	0.59	0.5520
Drop draws within 30 minutes of a filing	97,768	0.72	0.59	0.5520
Keep only draws on a filing day	3,276	9.24	1.49	0.1375
<i>Panel C: Filing-minus-placebo</i>				
Placebo definition		Gap (bps)	t	p
All placebo draws		8.14***	4.57	< 0.0001
Excluding draws on a filing day		8.40***	4.67	< 0.0001
Excluding draws on or next to a filing day		8.22***	4.49	< 0.0001

Table IA.12: Placebo draws that coincide with a real filing

Note: The sample (EXCL form types) includes all form types excluding 3/4/5/13D/13G accepted by EDGAR between September 2003 and December 2024 and the placebo events described in Section 5.1. Filings and placebo events are restricted to informative firm-quarters (Section 3.3). Pre- and post-acceptance returns are measured over 30-minute windows around the EDGAR acceptance timestamp of each filing and around the event time of each placebo event. Because the placebo draws a firm with replacement and a date uniformly among that firm’s eligible trading days, a draw can land on a day on which the firm actually filed. Panel A reports how often that happens, relative to the real trading-hour filings that cleared the CIK–CUSIP match and the overlap rule, so the same-day rate is a lower bound on the share of draws that fall on a day with EDGAR activity. Panel B re-forms the quintiles and re-estimates the placebo 5–1 after removing the coincident draws, and Panel C recomputes the filing-*minus*-placebo difference against each cleaned placebo. The t -statistics are Welch two-sample statistics; the difference tests the difference of the two 5–1s. To indicate significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

IA.G. Main Results under Alternative Event-Price Definitions

	Baseline	Alternative
<i>Panel A: Single sort and placebo tests, 5–1 (bps)</i>		
EXCL	8.86*** (6.78)	11.84*** (8.69)
Placebo	0.72 (0.59)	3.98*** (3.26)
Filing-minus-placebo	8.14*** (4.57)	7.86*** (4.30)
All form types	4.18*** (5.22)	6.46*** (7.60)
Scheduled	5.68** (2.44)	9.60*** (4.01)
Unscheduled	9.80*** (6.30)	12.56*** (7.69)
Form 8-K	15.52*** (5.49)	19.80*** (6.79)
Form 10-K	3.21 (0.60)	6.22 (1.16)
Form 10-Q	7.52** (2.20)	12.06*** (3.43)
<i>Panel B: After known hacking episodes and additional robustness tests, EXCL 5–1 (bps)</i>		
Filings on or after 1 March 2015	5.58*** (2.85)	8.03*** (3.79)
Filings on or after 1 October 2020	7.49*** (3.19)	8.59*** (3.42)
90-minute window	10.75*** (3.76)	12.43*** (4.35)
<i>Panel C: Alternative explanations, slope of Q_{30}^{post} on Q_{30}^{pre} (standardized)</i>		
No controls	0.0661*** (5.04)	0.0863*** (6.00)
All controls	0.0707*** (4.42)	0.0900*** (5.14)
<i>Panel D: Pre-event trading activity</i>		
Abnormal volume double sort, 5–1 of 5–1 (bps)	16.95*** (4.47)	19.07*** (4.74)
QID^{res} double sort, 5–1 of 5–1 (bps)	14.27** (2.24)	17.96*** (2.65)
Lee–Ready order imbalance, 5–1 of X_{LR}	0.2458*** (75.56)	0.2526*** (76.71)

Table IA.13: Main results under the baseline and the alternative event price

Note: Each column re-runs the exhibit under a different definition of the event price. All rows use filings in informative firm-quarters (Section 3.3). The baseline is the definition of Section 3.2. The alternative starts the post-acceptance return at $[-3, 0)$ rather than $[0, 3)$, where the pre-acceptance return ends. Entries are point estimates with the t -statistic of the summarized exhibit in parentheses: Welch two-sample statistics for the sort, placebo (as constructed in Section 5.1), double-sort, and Lee–Ready rows, and two-way firm-and-month clustered statistics for the panel rows. Sorts and double sorts are in basis points, the panel slopes are standardized, and the Lee–Ready row is a share. The alternative requires a different non-missing three-minute price, so it covers a slightly different number of filings. For the 90-minute window, the alternative pre-acceptance return is the baseline one, from $[-90, -87)$ to $[-3, 0)$, and the post-acceptance return runs from $[-3, 0)$ to $[87, 90)$. To indicate significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

IA.H. Quote-Midpoint Prices

Event-price definition	N	Filings		Placebo		Difference	
		5–1 (bps)	<i>t</i>	5–1 (bps)	<i>t</i>	(bps)	<i>t</i>
Baseline	115,455	5.87***	5.64	0.09	0.12	5.77***	4.36
Alternative	115,455	9.66***	8.71	4.14***	4.77	5.53***	3.92

Table IA.14: Quote-midpoint prices

Note: The sample (EXCL form types) includes all form types excluding 3/4/5/13D/13G accepted by EDGAR between September 2003 and December 2024 and the placebo events described in Section 5.1. Filings and placebo events are restricted to informative firm-quarters (Section 3.3). Pre- and post-acceptance returns are measured over 30-minute windows around the EDGAR acceptance timestamp of each filing and around the event time of each placebo event. The returns are recomputed from NBBO midpoints, averaged over the same four three-minute edge windows as in Section 3.2, rather than from transaction prices. Because a midpoint exists whenever a quote is outstanding, the midpoint sample is larger than the transaction-price sample and includes more thinly traded filings. The baseline and alternative rows use the baseline and alternative event prices defined in Section 3.2. The *t*-statistics are Welch two-sample statistics; the difference tests the difference of the two 5–1s. To indicate significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

IA.I. Single Sort, Filings on or after 1 March 2015

Q_{30}^{pre}	bin	Mean Q_{30}^{post} (bps)	t	Std. err. (bps)
	5	1.61	1.17	1.38
	4	-0.41	-0.67	0.60
	3	0.38	0.67	0.56
	2	-0.92	-1.40	0.66
	1	-3.96***	-2.86	1.38
5–1		5.58***	2.85	1.95
Total Observations N = 24,459				

Table IA.15: Single sort, EXCL, 30-minute windows, on or after 1 March 2015

Note: The sample (EXCL form types) includes all form types excluding 3/4/5/13D/13G accepted by EDGAR from 1 March 2015, after the SEC closed the FTP/PDS dissemination defect, through December 2024. Filings are restricted to informative firm-quarters (Section 3.3). Pre- and post-acceptance returns are measured over 30-minute windows around the EDGAR acceptance timestamp of each filing. Quintile means report one-sample t -statistics, and the 5–1 reports the Welch two-sample statistic. To indicate significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

IA.J. Single Sort, Filings on or after 1 October 2020

Q_{30}^{pre}	bin	Mean Q_{30}^{post} (bps)	t	Std. err. (bps)
	5	1.22	0.82	1.48
	4	-0.57	-0.68	0.85
	3	0.57	0.71	0.80
	2	-0.44	-0.43	1.02
	1	-6.26***	-3.45	1.82
5–1		7.49***	3.19	2.35
Total Observations N = 11,142				

Table IA.16: Single sort, EXCL, 30-minute windows, on or after 1 October 2020

Note: The sample (EXCL form types) includes all form types excluding 3/4/5/13D/13G accepted by EDGAR from 1 October 2020, after the alleged filing-agent hacking window, through December 2024. Filings are restricted to informative firm-quarters (Section 3.3). Pre- and post-acceptance returns are measured over 30-minute windows around the EDGAR acceptance timestamp of each filing. Quintile means report one-sample t -statistics, and the 5–1 reports the Welch two-sample statistic. To indicate significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

IA.K. Self-Filed and Agent-Filed Forms

	N	5–1 (bps)	<i>t</i>
Self-filed	14,203	11.76***	4.45
Agent-filed	54,366	7.89***	5.28
Self minus agent		4.66	1.54

Table IA.17: Self-filed versus agent-filed filings

Note: The sample (EXCL form types) includes all form types excluding 3/4/5/13D/13G accepted by EDGAR between September 2003 and December 2024. Filings are restricted to informative firm-quarters (Section 3.3). Pre- and post-acceptance returns are measured over 30-minute windows around the EDGAR acceptance timestamp of each filing. Following [Barger, Clifford, and Qiu \(2025\)](#), a filing is self-filed when the CIK in the first ten digits of its accession number, which identifies the entity that transmitted it, equals the subject company’s own CIK, and agent-filed otherwise. The classification is a proxy: a parent transmitting on behalf of a subsidiary, or an agent that is itself a registrant, is misclassified. The first two rows form pre-acceptance-return quintiles within each group and report Welch two-sample *t*-statistics. The third row is the interaction from a pooled sort, which tests whether the 5–1 differs between the two groups. To indicate significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

IA.L. Single Sort, 90-Minute Window

Q_{90}^{pre}	bin	Mean Q_{90}^{post} (bps)	t	Std. err. (bps)
	5	4.83**	2.52	1.92
	4	1.32	1.30	1.02
	3	-0.13	-0.13	1.02
	2	-3.61***	-3.22	1.12
	1	-5.92***	-2.79	2.12
5–1		10.75***	3.76	2.86
Total Observations N = 34,780				

Table IA.18: Single sort, EXCL, 90-minute windows

Note: The sample (EXCL form types) includes all form types excluding 3/4/5/13D/13G accepted by EDGAR between September 2003 and December 2024. Filings are restricted to informative firm-quarters (Section 3.3). Pre- and post-acceptance returns are measured over 90-minute windows around the EDGAR acceptance timestamp of each filing. Quintile means report one-sample t -statistics, and the 5–1 reports the Welch two-sample statistic. To indicate significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.